

MACHINE LEARNING FOR SIGNAL PROCESSING

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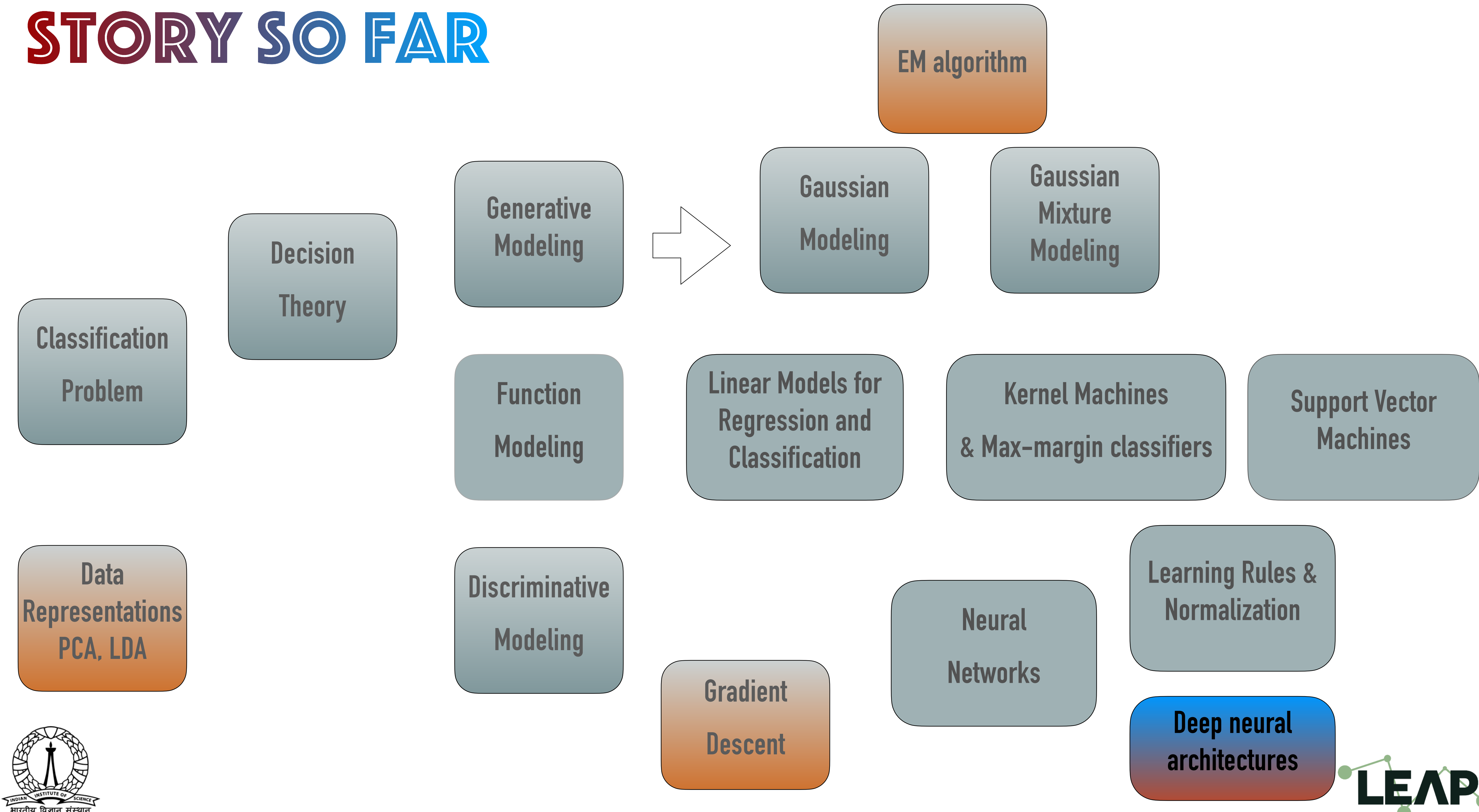
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Viveka Salinamakki, Varada R.

LEAP lab, Electrical Engineering, Indian Institute of Science
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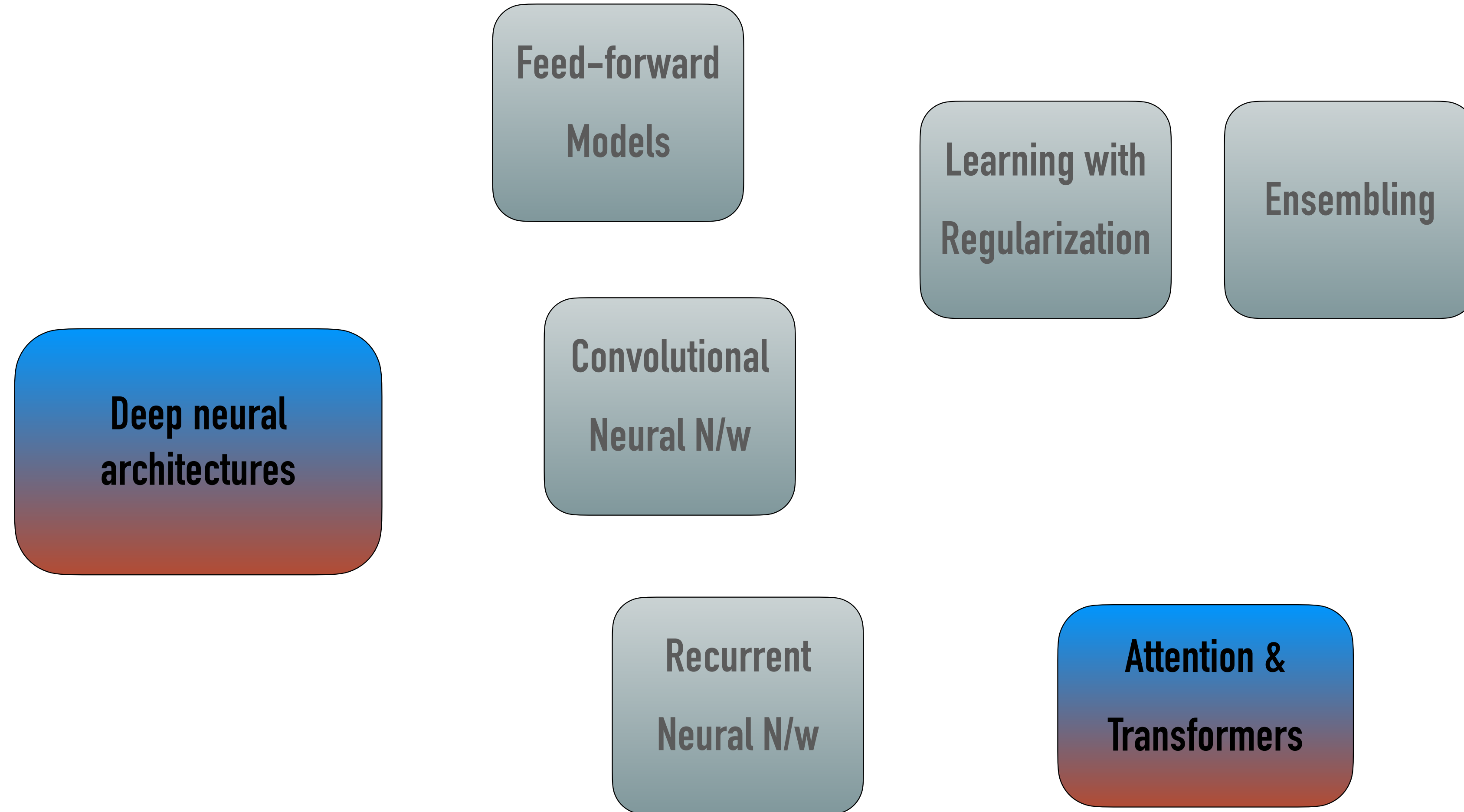
<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>



STORY SO FAR



STORY SO FAR



RECURRENT NEURAL NETWORKS

INTRODUCTION

- The standard DNN/CNN paradigms
 - * $(\mathbf{x}, \mathbf{y})$ - ordered pair of data vectors/images ($\mathbf{x}$) and target ($\mathbf{y}$)
- Moving to sequence data
 - * $(\mathbf{x}(t), \mathbf{y}(t))$ where this could be sequence to sequence mapping task.
 - * $(\mathbf{x}(t), \mathbf{y})$ where this could be a sequence to vector mapping task.
 - ◆ Input features / output targets are correlated in time.
 - ◆ Unlike standard models where each pair is independent.
 - ◆ Need to model dependencies in the sequence over time.

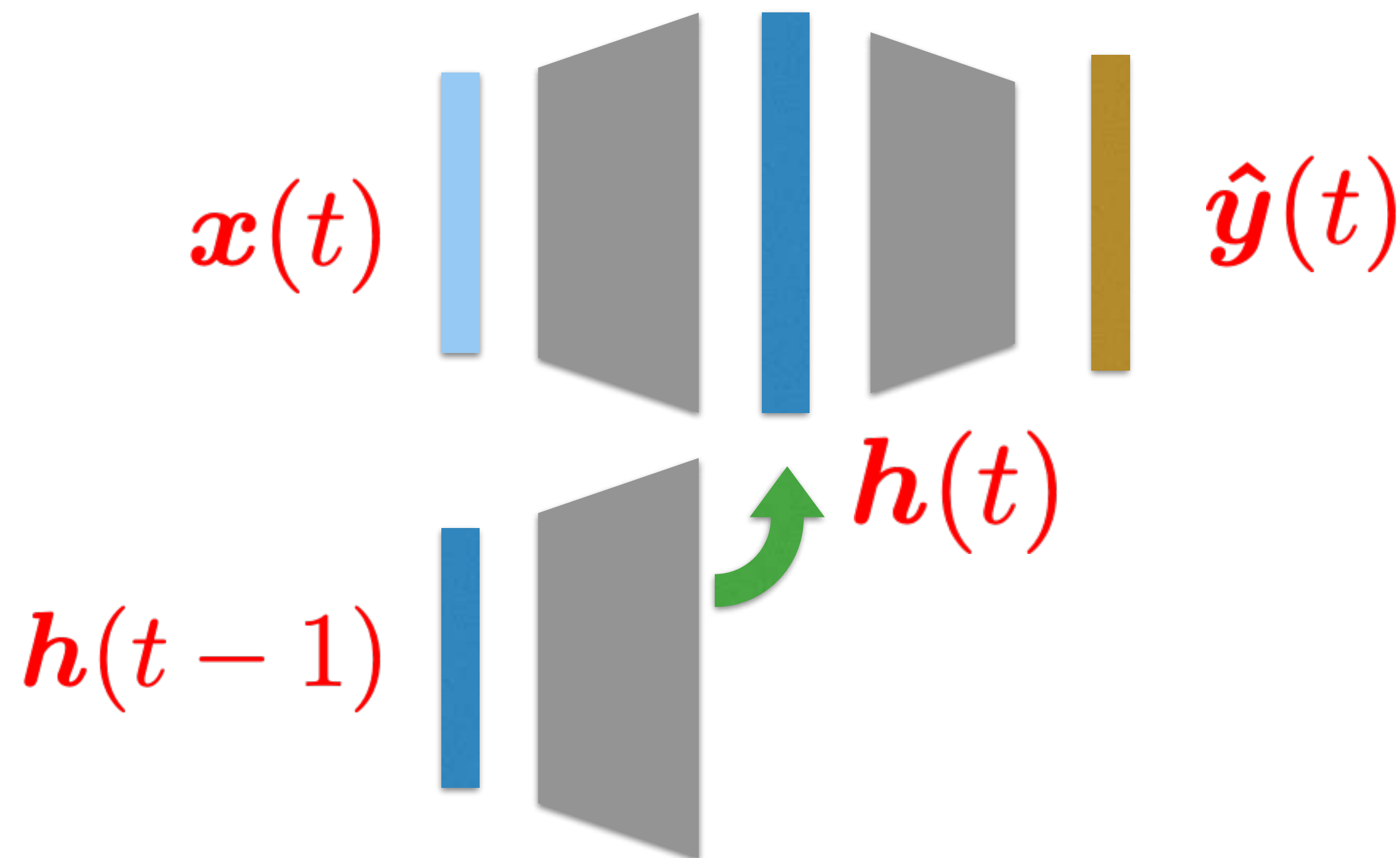
WHY DO WE NEED RECURRENT MODELS

- An interesting subset of this problem is where the input alone is a time series or have different indices $x(t), y$
- Examples
 - ◆ Text sequences
 - ◆ Speech and audio
 - ◆ Video sequences
 - ◆ ECG/EEG data
 - ◆ Wearable sensor data

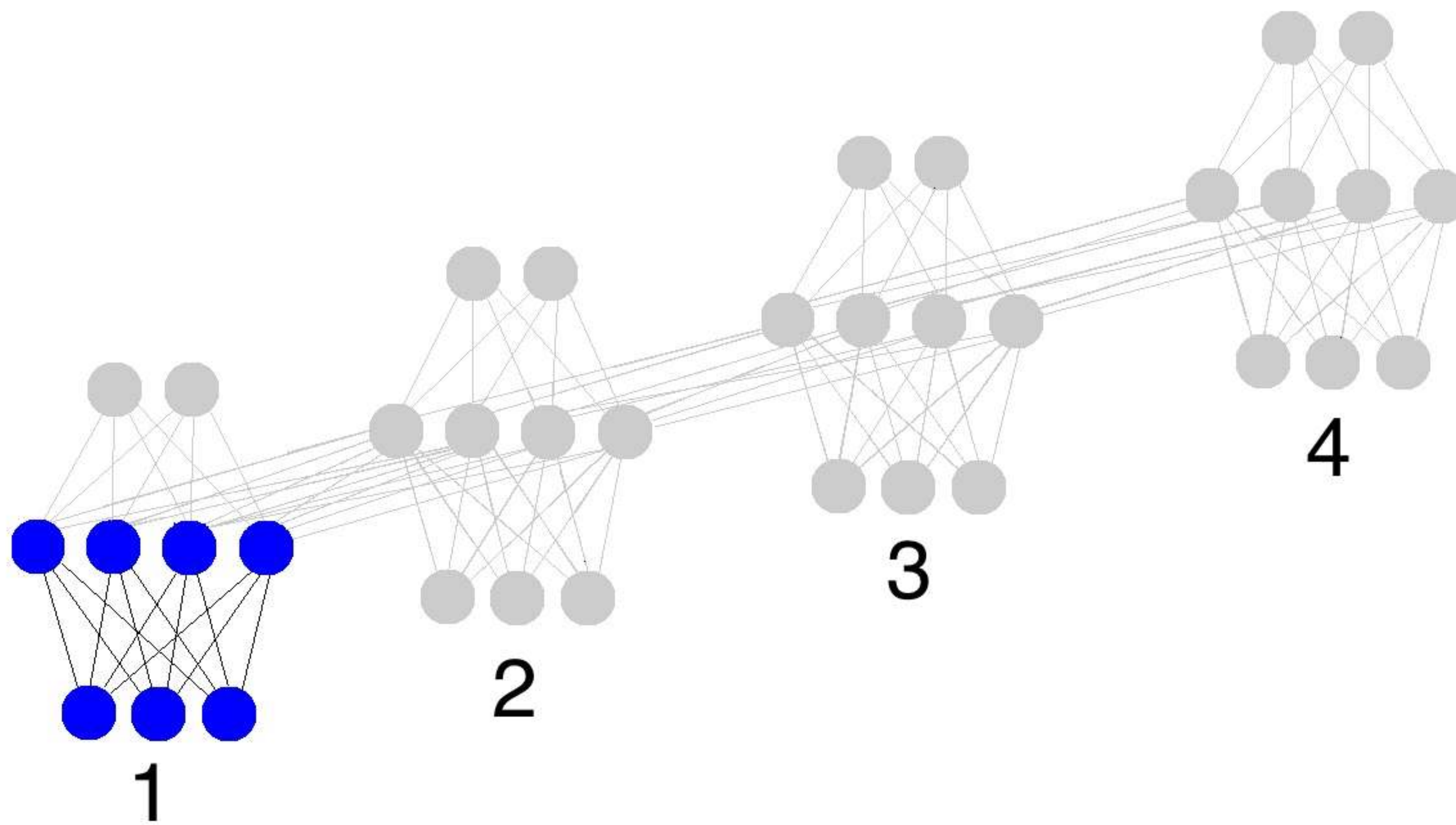
FIRST ORDER RECURRENCE – HIDDEN LAYER

- Making the hidden layer a function of the previous outputs from the hidden layer along with the input

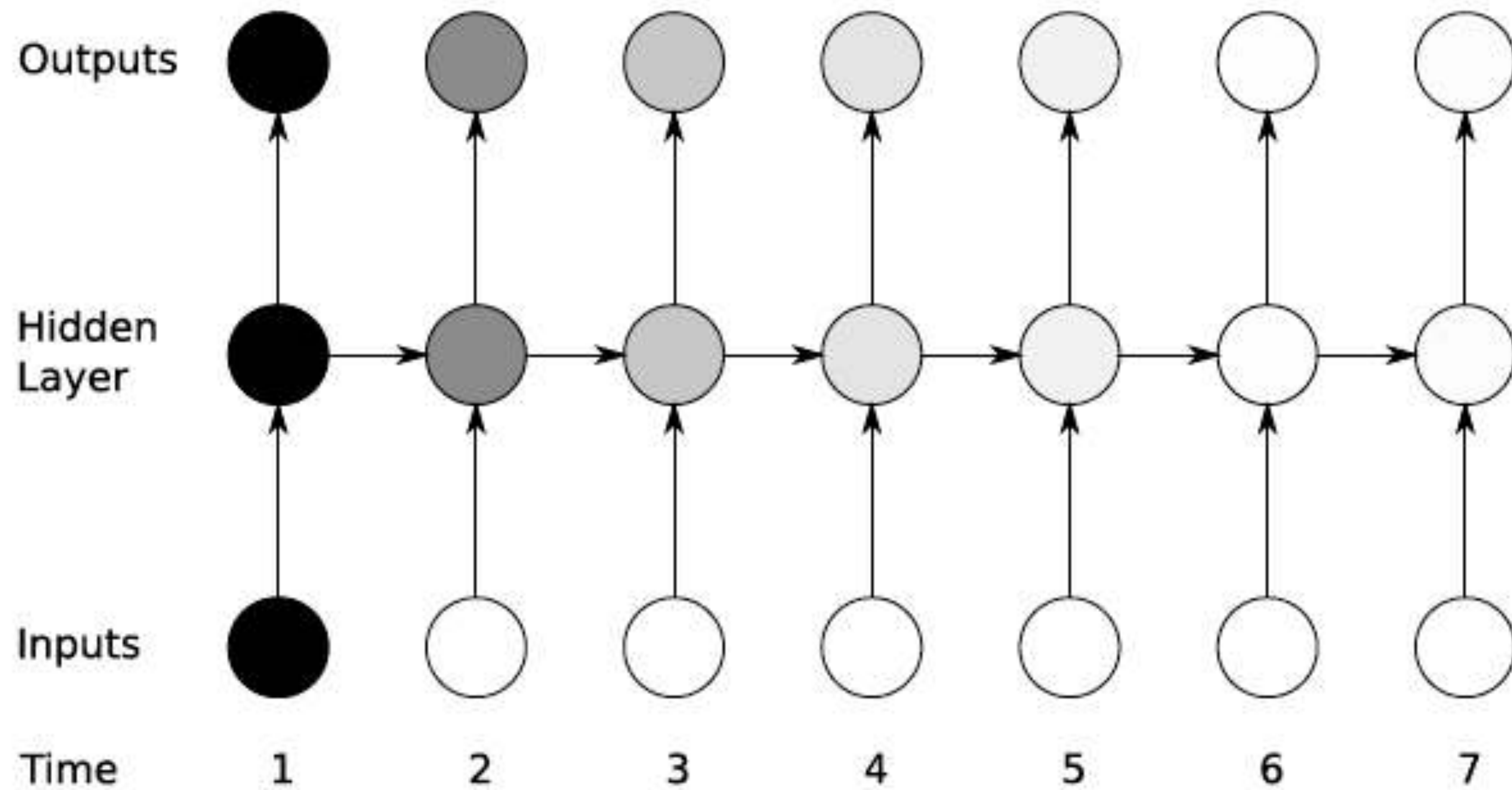
$$\mathbf{h}(t) = f(\mathbf{h}(t-1), \mathbf{x}(t))$$



BACK PROPAGATION THROUGH TIME

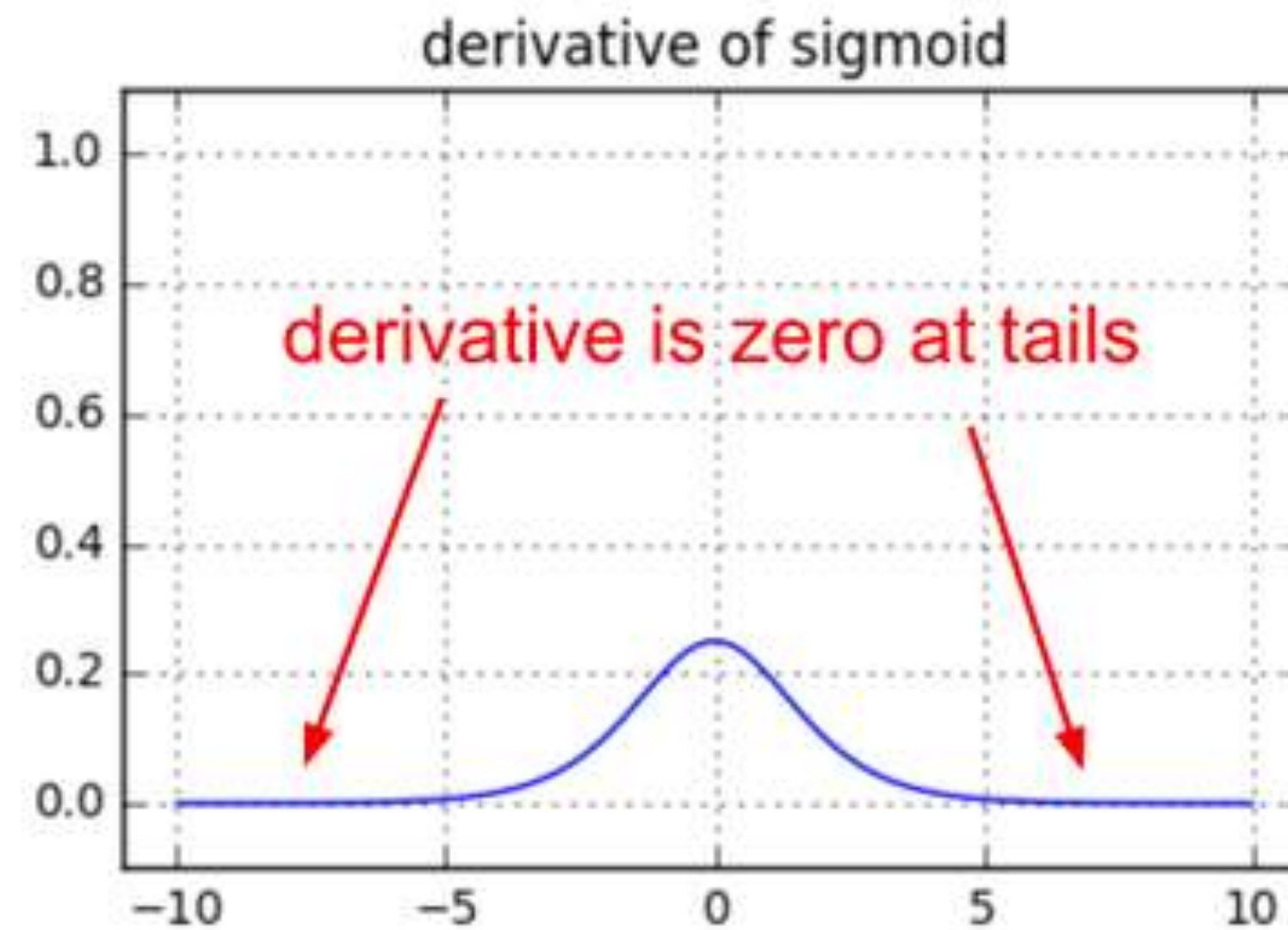
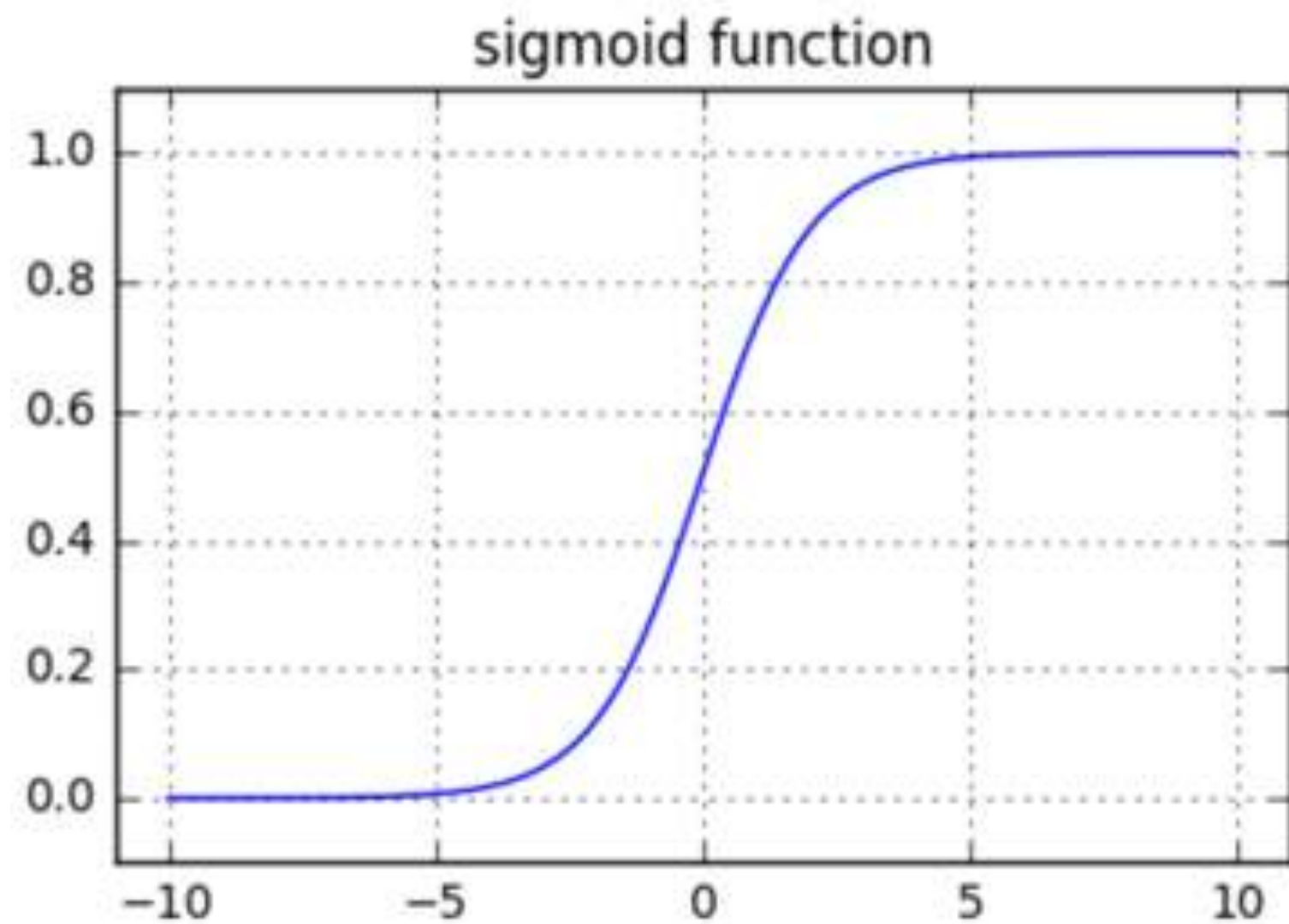


LONG-TERM DEPENDENCY ISSUES

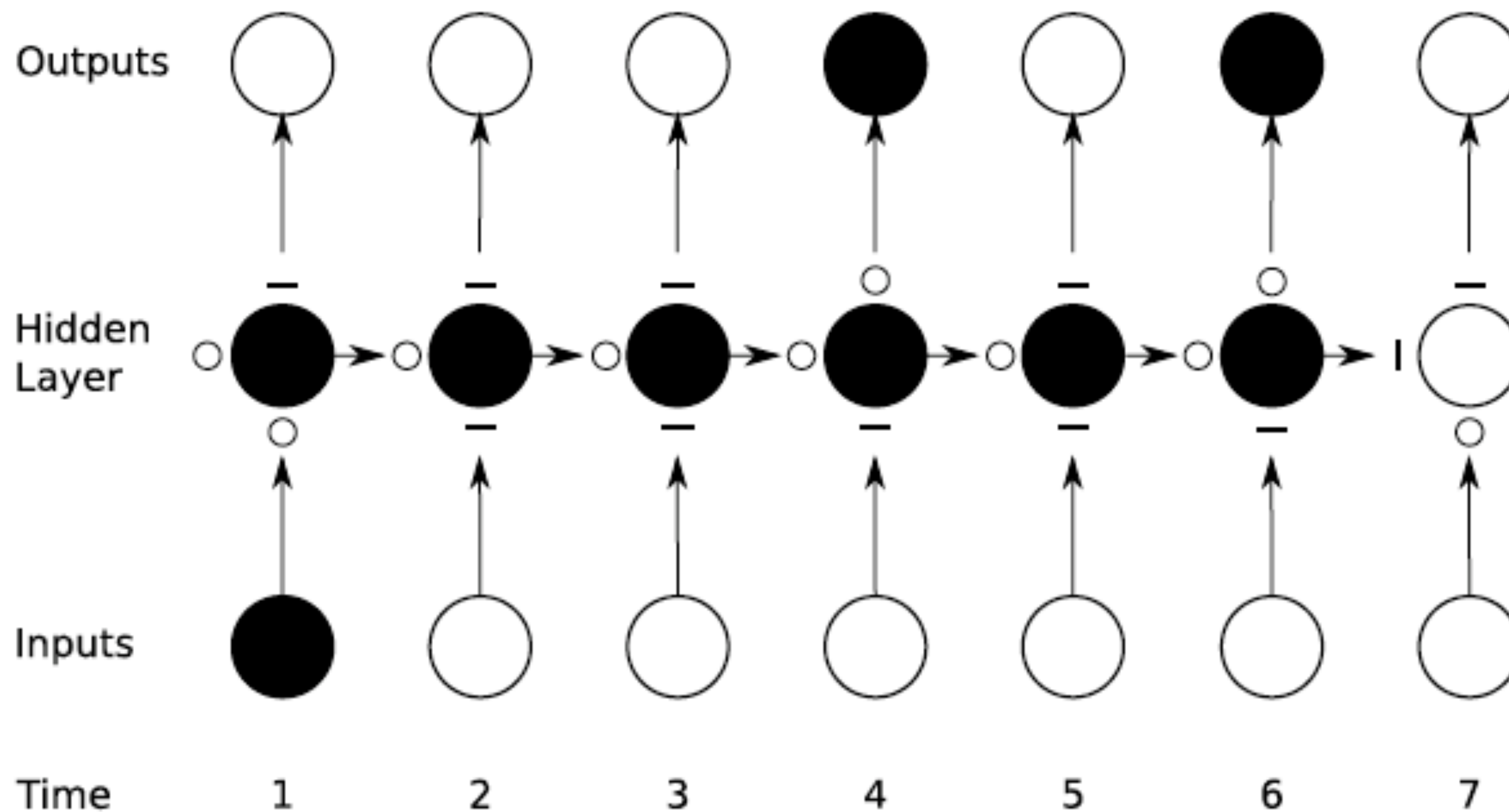


LONG-TERM DEPENDENCY ISSUES

- ◆ Gradients tend to vanish or explode



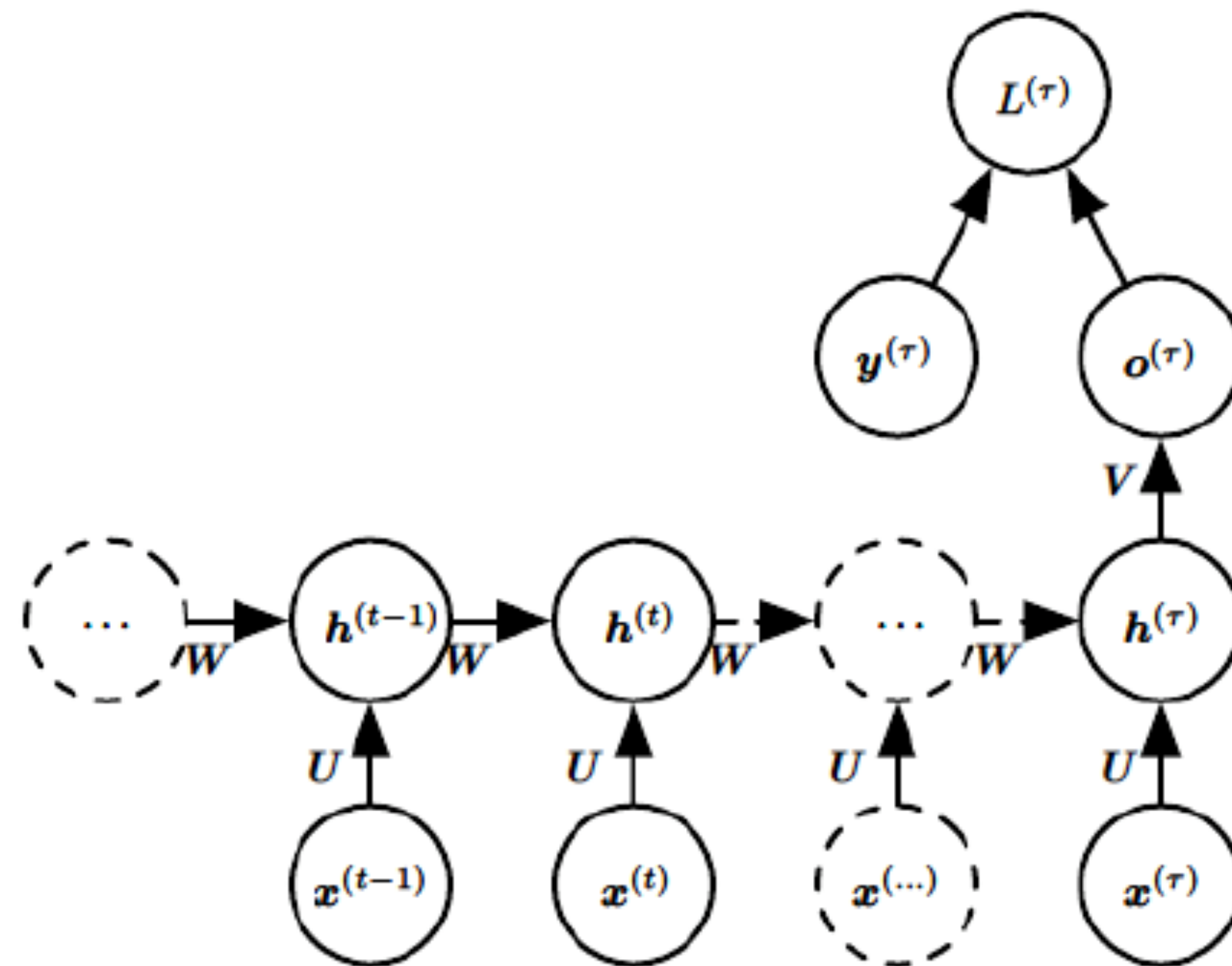
LONG SHORT TERM MEMORY (LSTM) IDEA



MODELING QUESTIONS

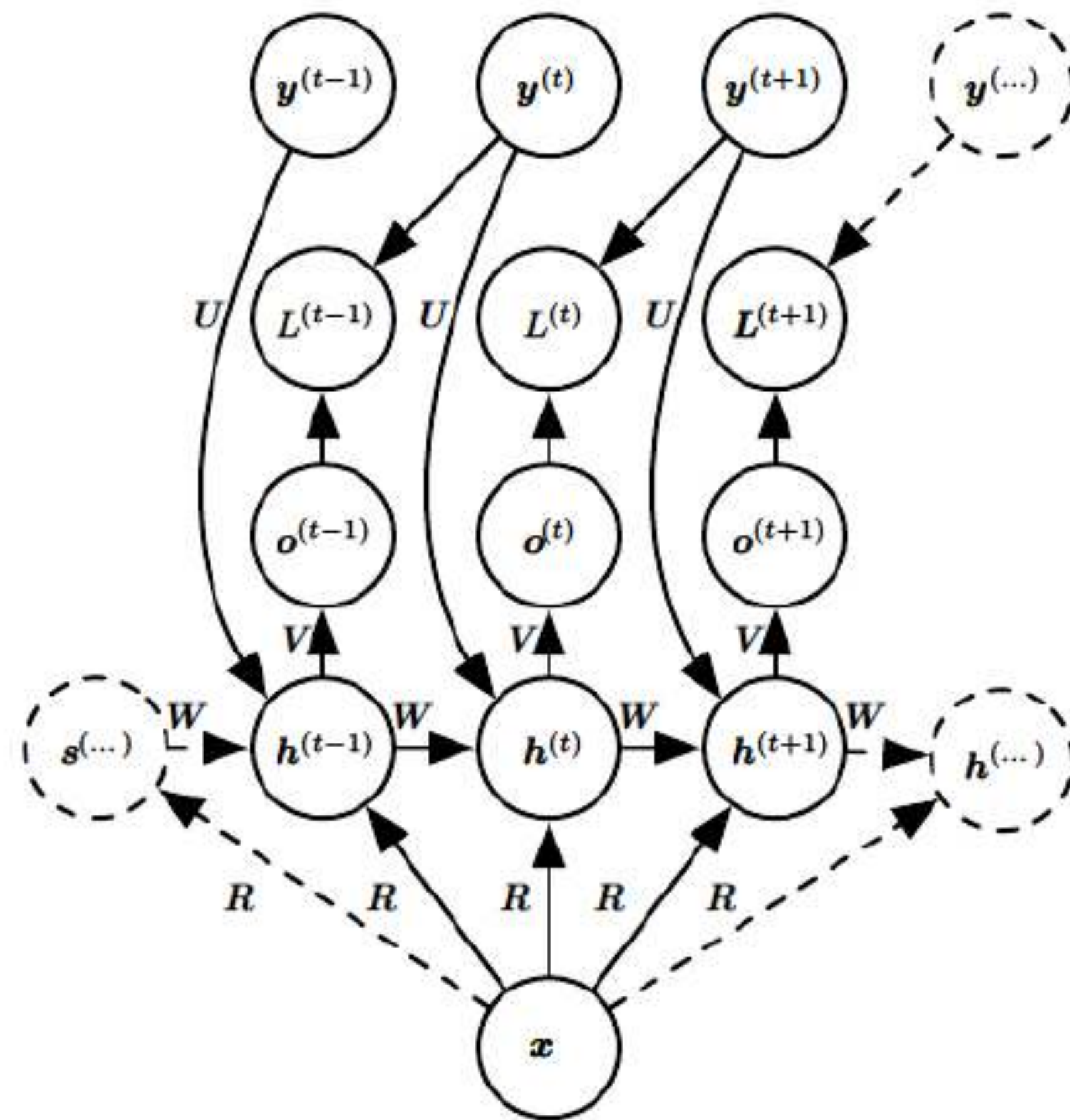
- How can we make adaptable gates with neural networks
 - * How can we make gates dependent on the data itself.
 - ◆ Gates can be implemented as neural layers with sigmoidal outputs ?
 - Sigmoids can approximate 0-1 functions
 - ✓ Modulate the gate output with inputs, hidden layer outputs or outputs

Recurrent Networks



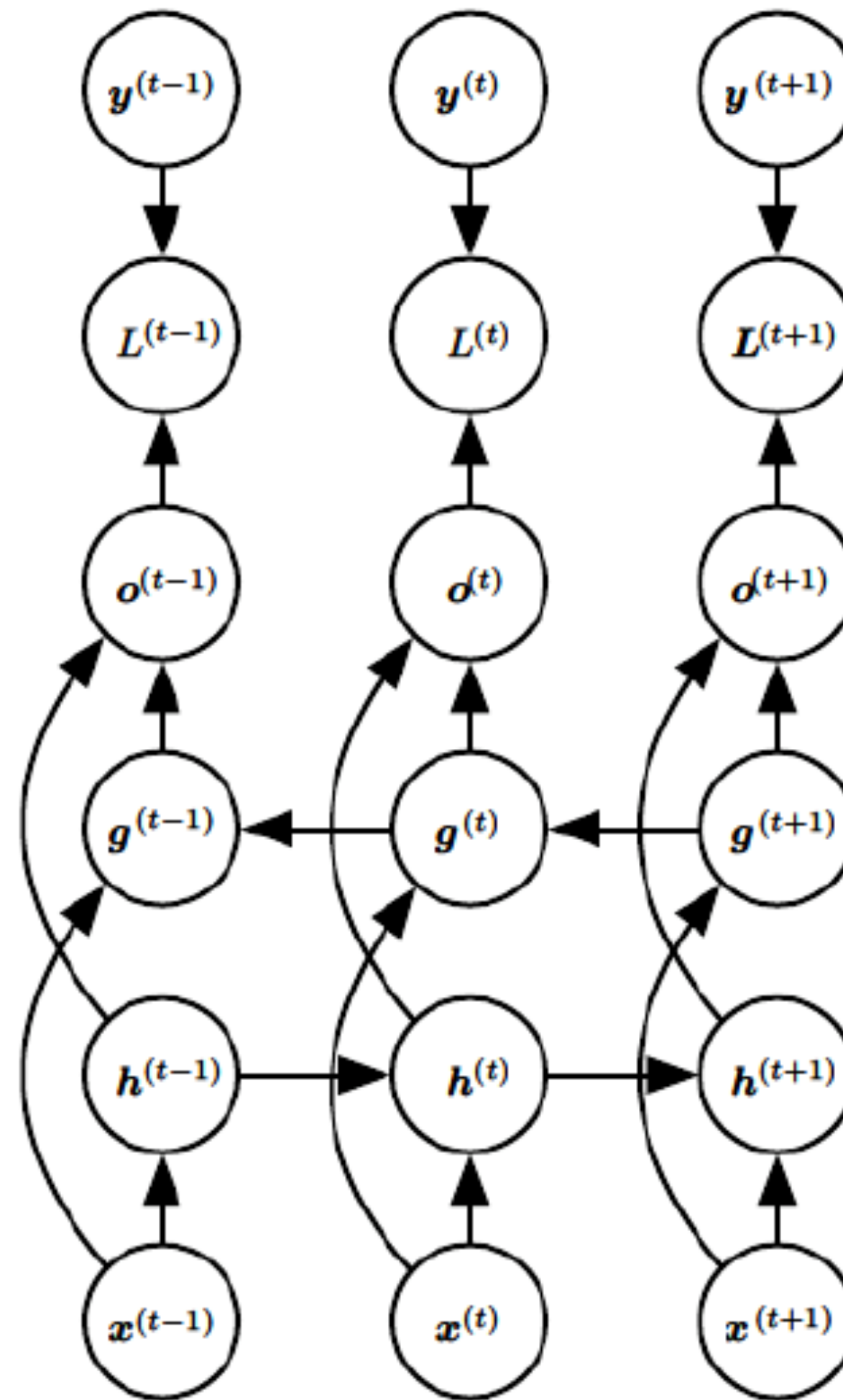
**Multiple Input
Single Output**

Recurrent Networks



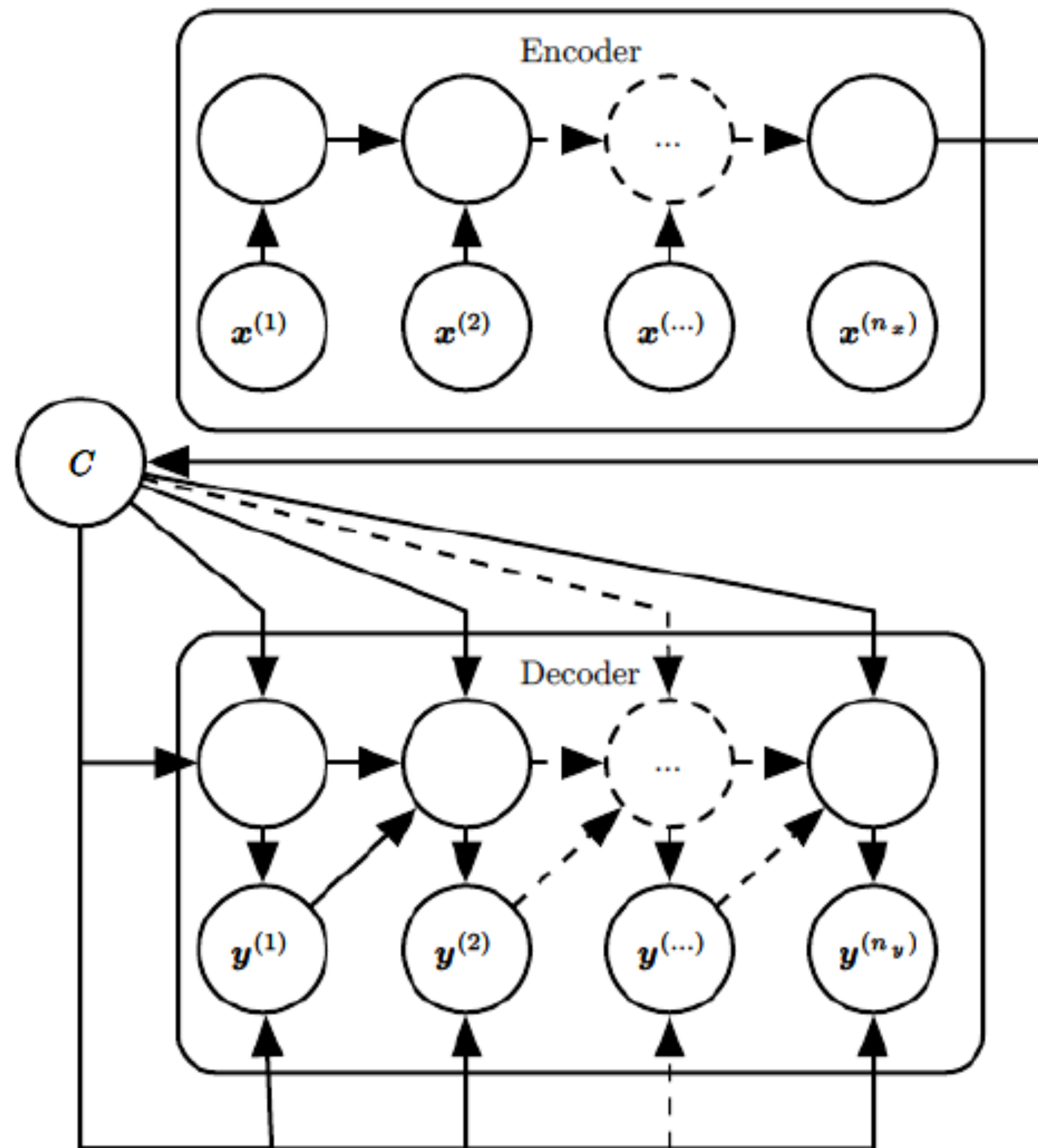
**Single Input
Multiple Output**

Recurrent Networks



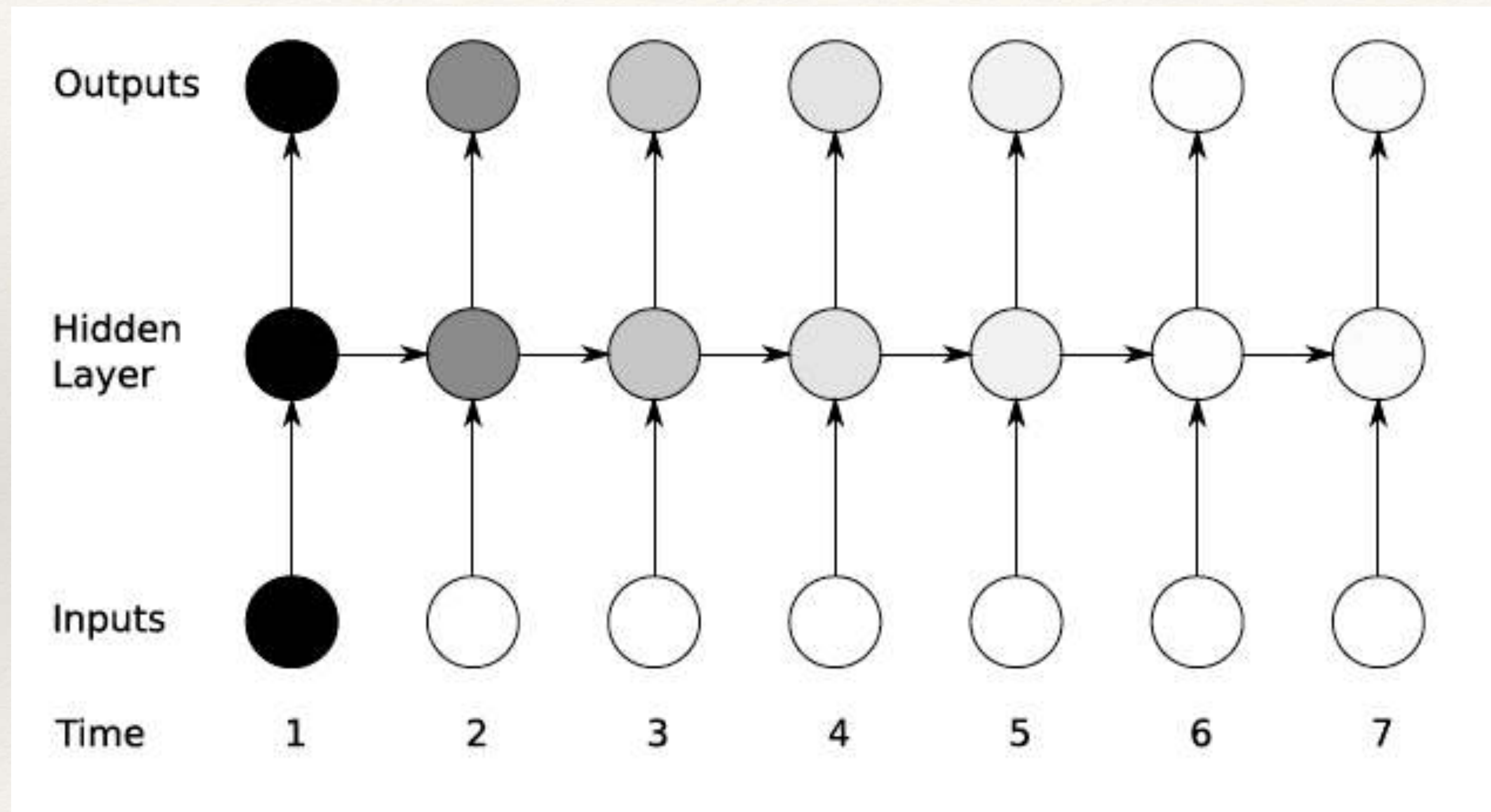
**Bi-directional
Networks**

Recurrent Networks

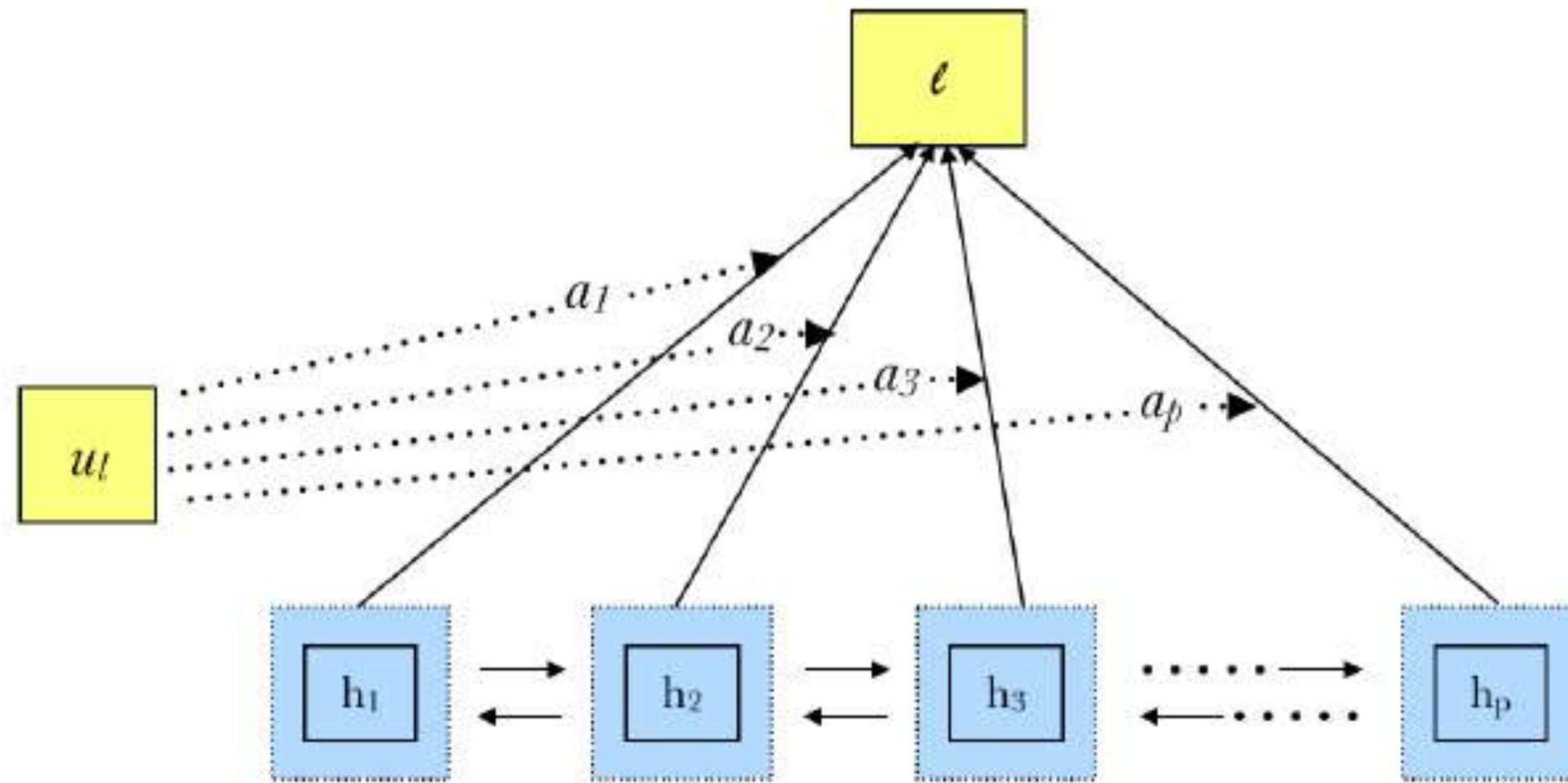


**Sequence to
Sequence
Mapping Networks**

Long-term Dependency Issues



Attention in LSTM Networks



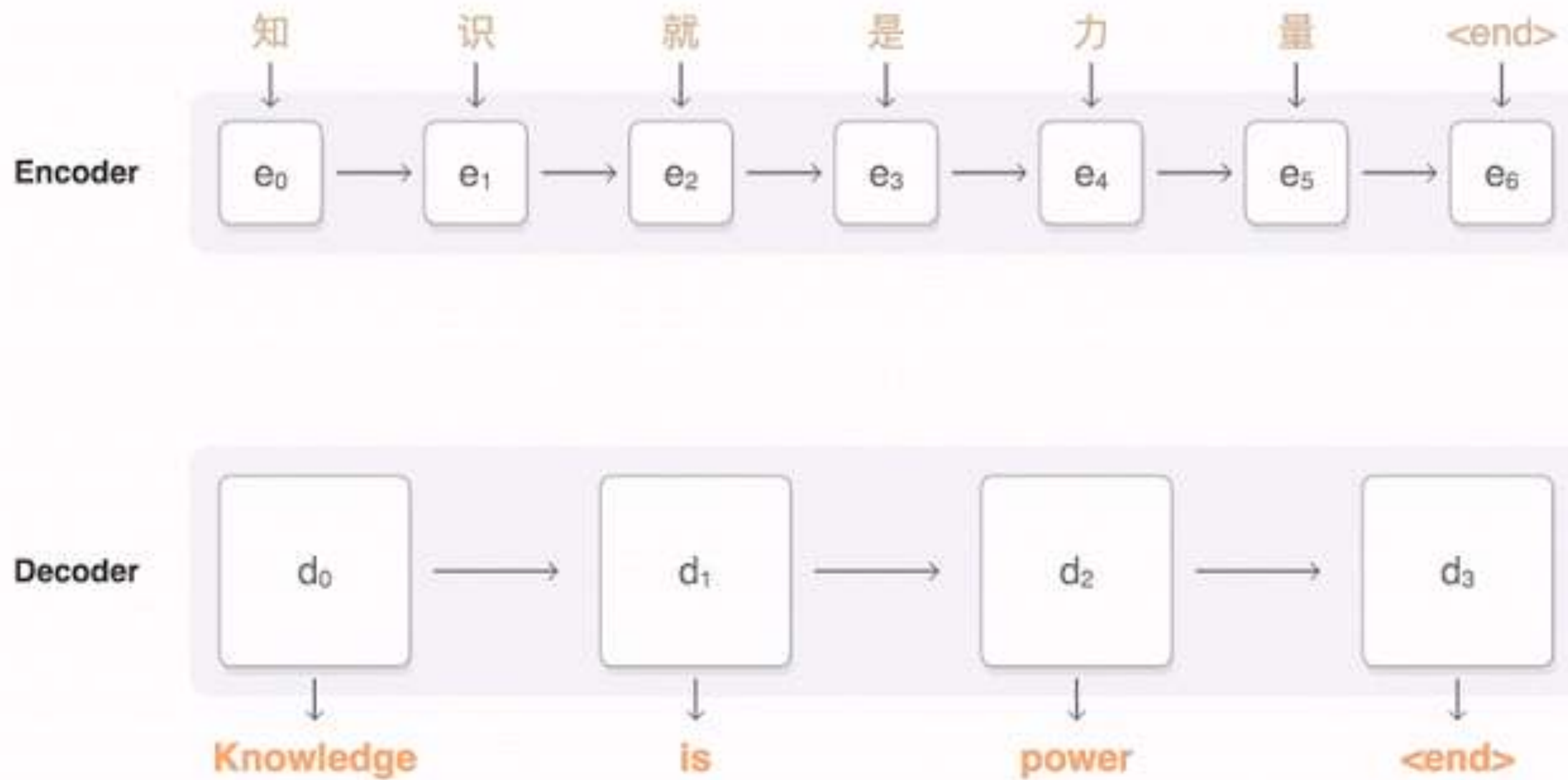
$$\mathbf{u}_t = \tanh(\mathbf{W}_l \mathbf{h}_t + \mathbf{b}_l)$$

$$a_t = \frac{\exp(\mathbf{u}_t^T \mathbf{u}_l)}{\sum_t \exp(\mathbf{u}_t^T \mathbf{u}_l)}$$

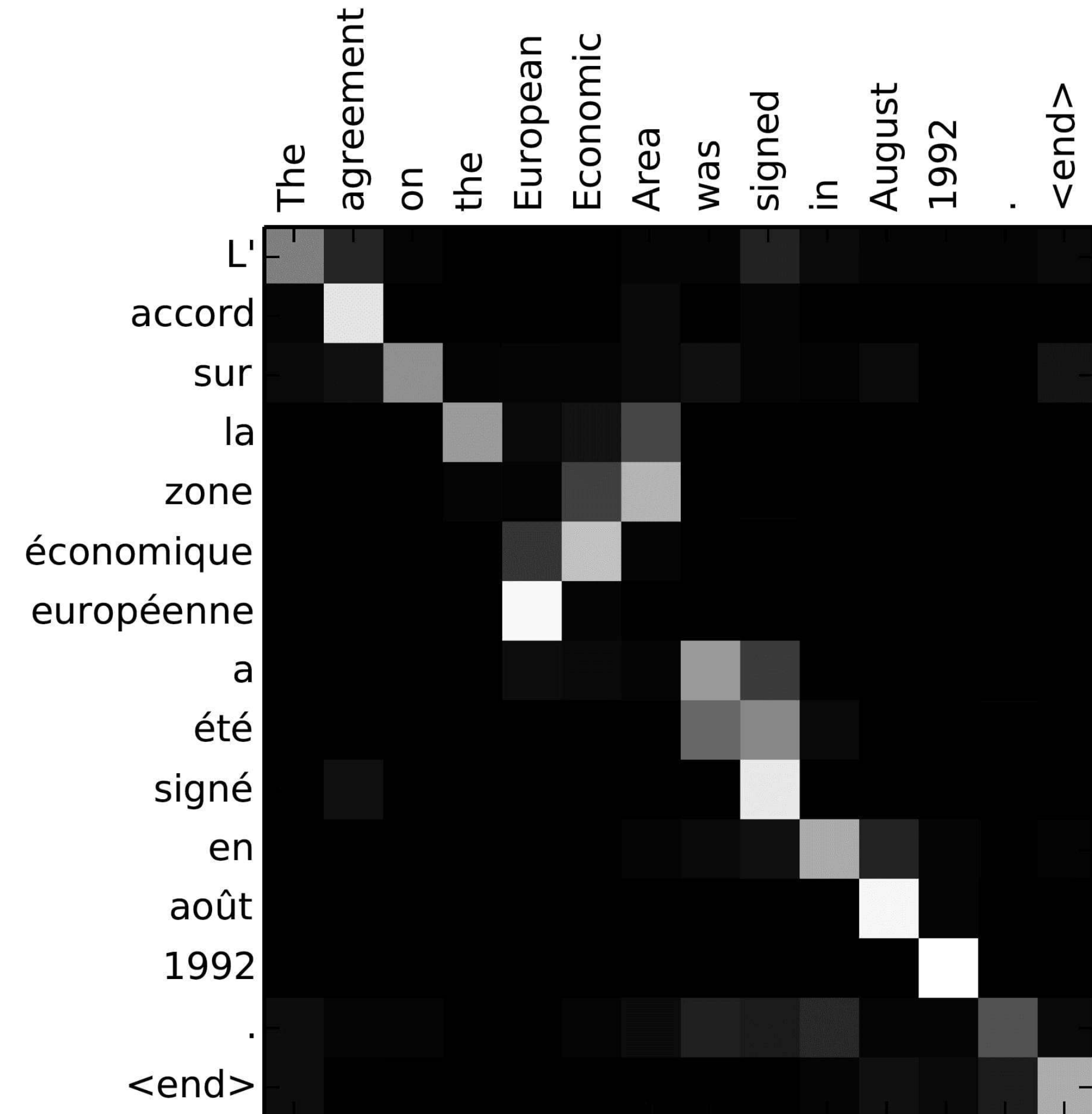
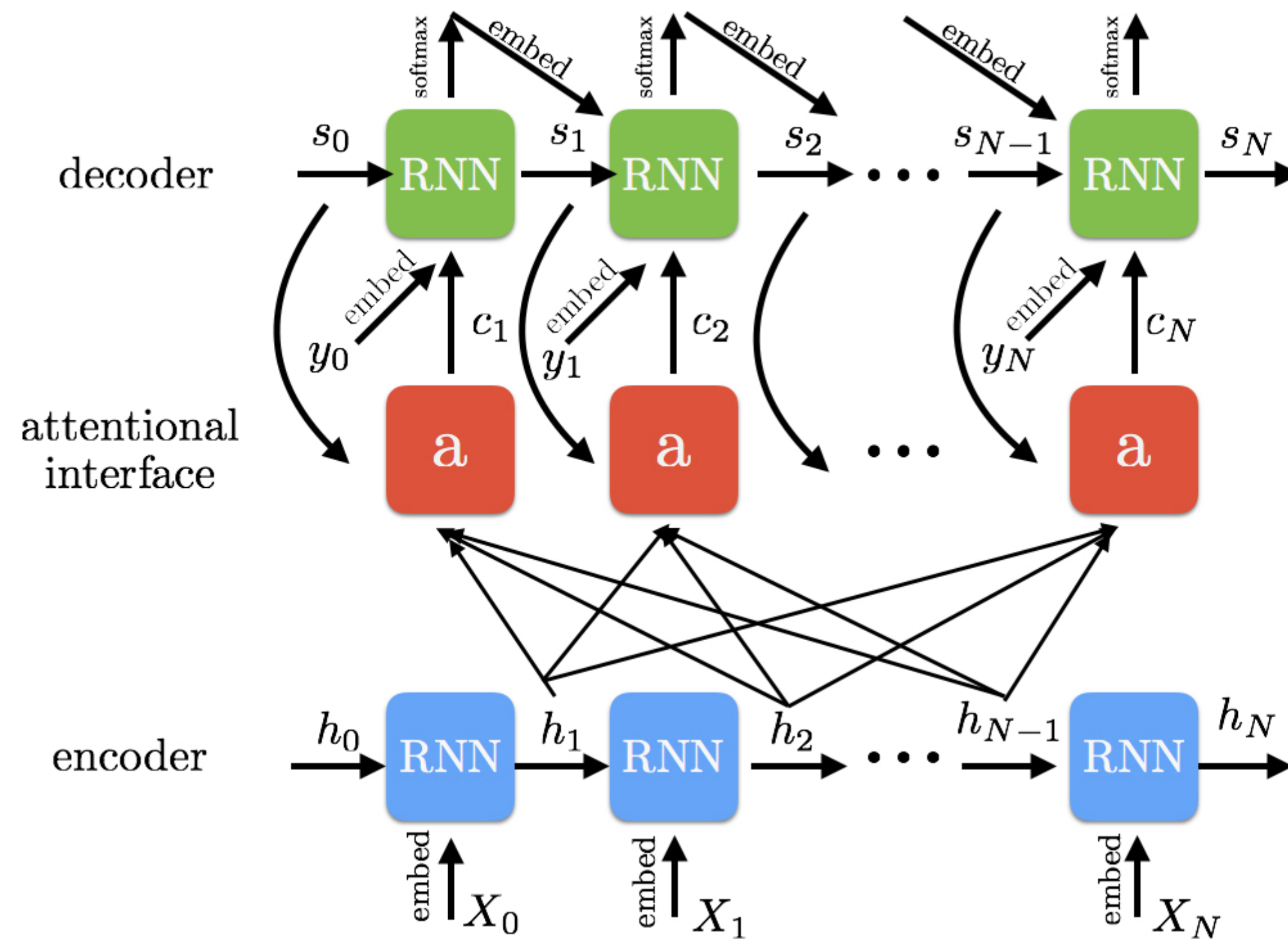
$$\mathbf{l} = \sum_t a_t \mathbf{h}_t$$

- ❖ Attention allows a mechanism to add relevance
- ❖ Certain regions of the audio have more importance than the rest for the task at hand.

Encoder - Decoder Networks with Attention

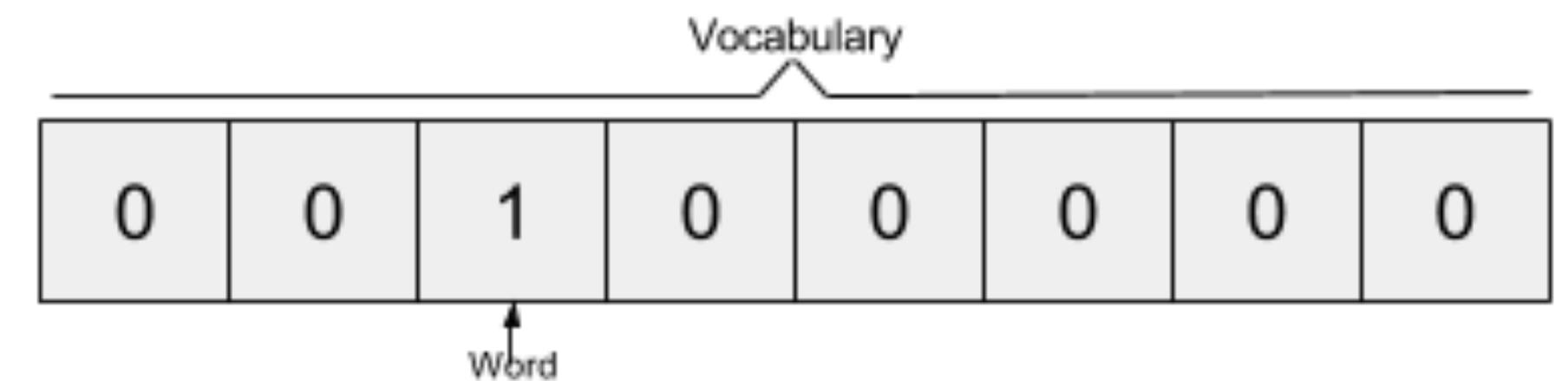


Self-Attention Models



Natural language processing - representations of text

- ❖ Converting text into fixed representations in a vector space.



- ❖ Using one-hot vectors is really high dimensional.
- ❖ Need a concise word representations that embeds semantics.

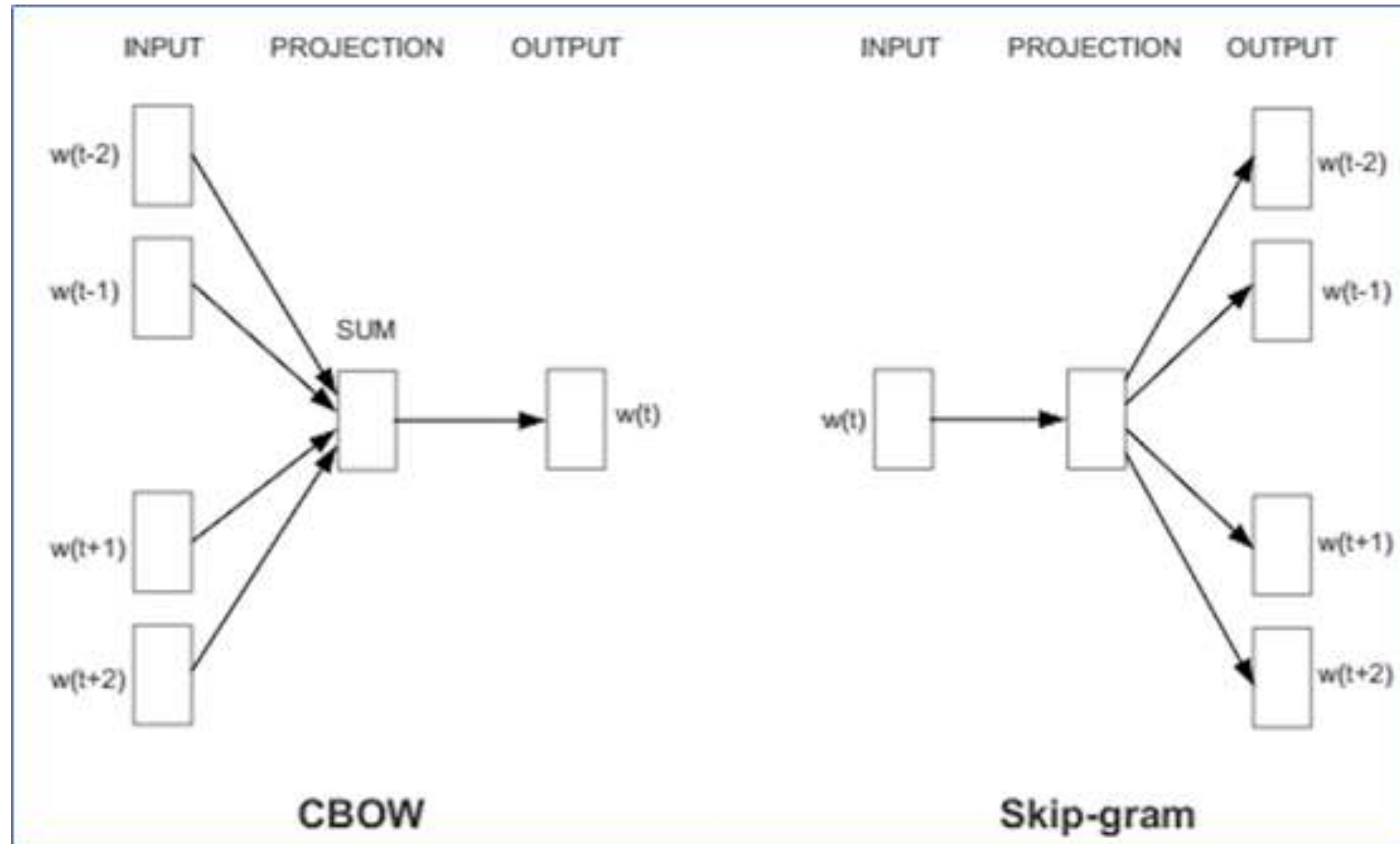
word2vec models as text representations

Source Text	Training Samples
The quick brown fox jumps over the lazy dog. The quick brown →	(the, quick) (the, brown)
The quick brown fox jumps over the lazy dog. The quick brown fox →	(quick, the) (quick, brown) (quick, fox)
The quick brown fox jumps over the lazy dog. The quick brown fox jumps →	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
The quick brown fox jumps over the lazy dog. The quick brown fox jumps over →	(fox, quick) (fox, brown) (fox, jumps) (fox, over)

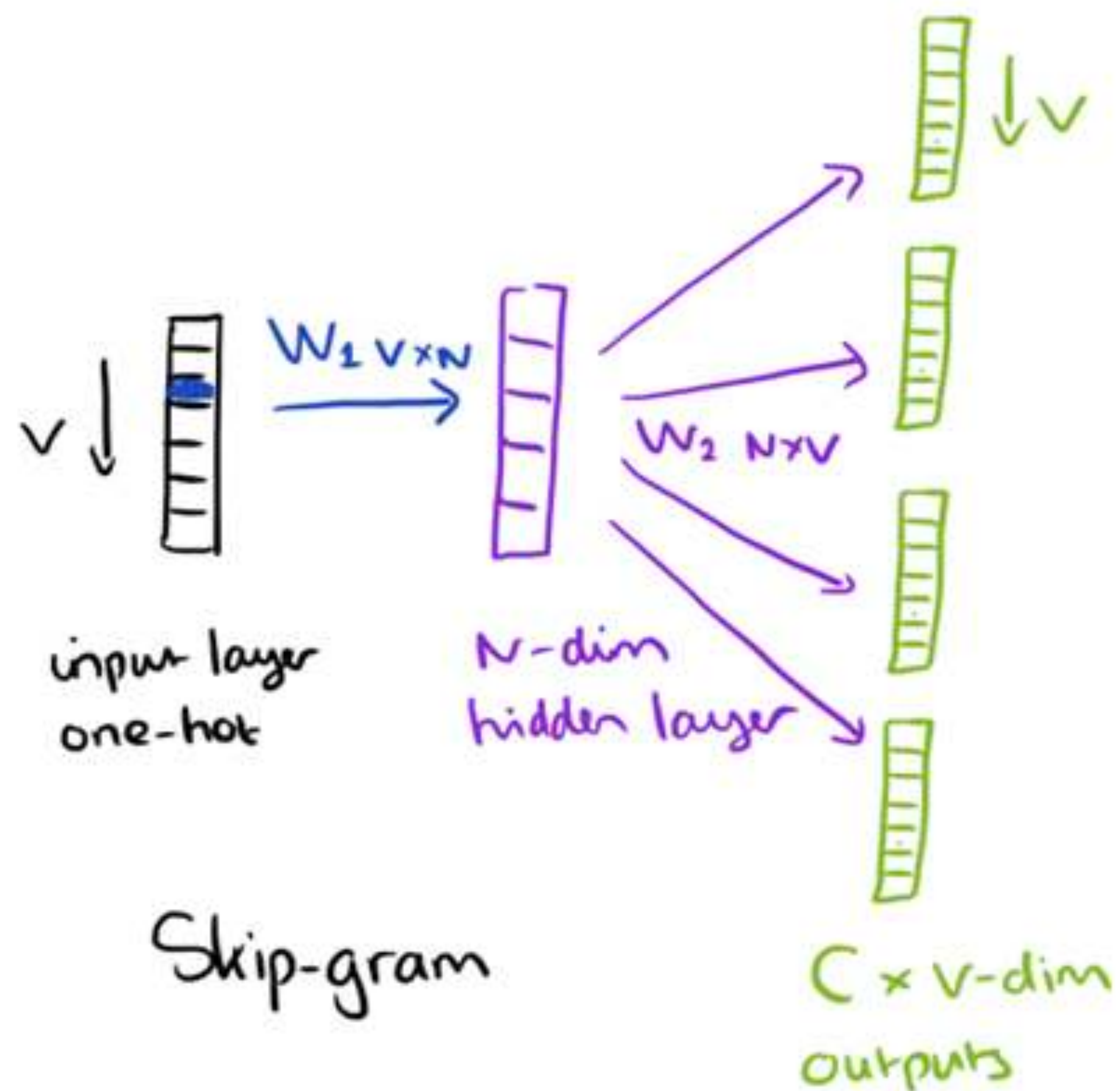
word2vec representations



word2vec models as text representations



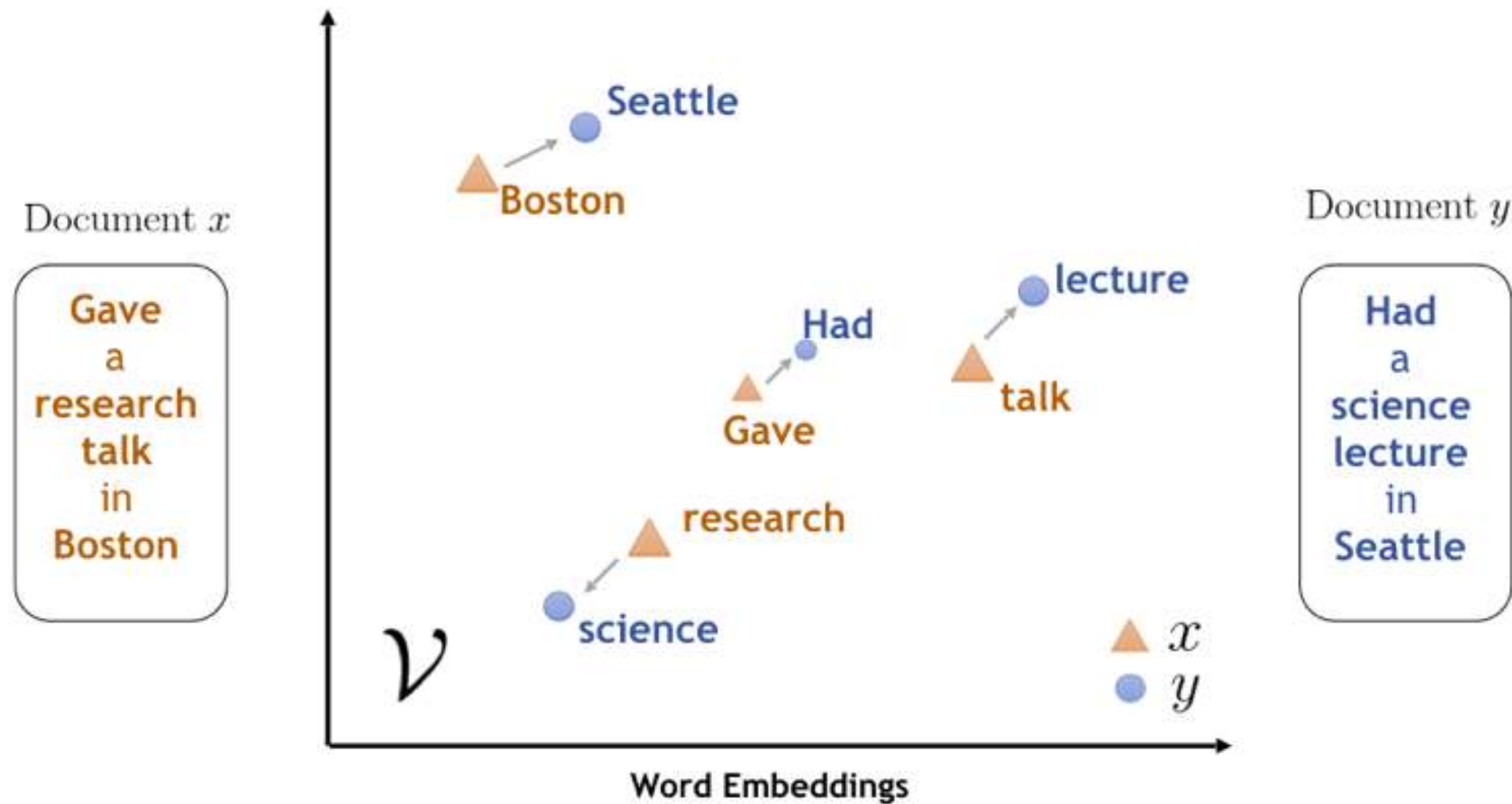
word2vec representations



$$\begin{array}{c} \text{input} \\ 1 \times V \end{array} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{array}{c} W_1 \\ V \times N \end{array} \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{bmatrix} = \begin{array}{c} \text{hidden} \\ 1 \times N \end{array} \begin{bmatrix} e & f & g & h \end{bmatrix}$$

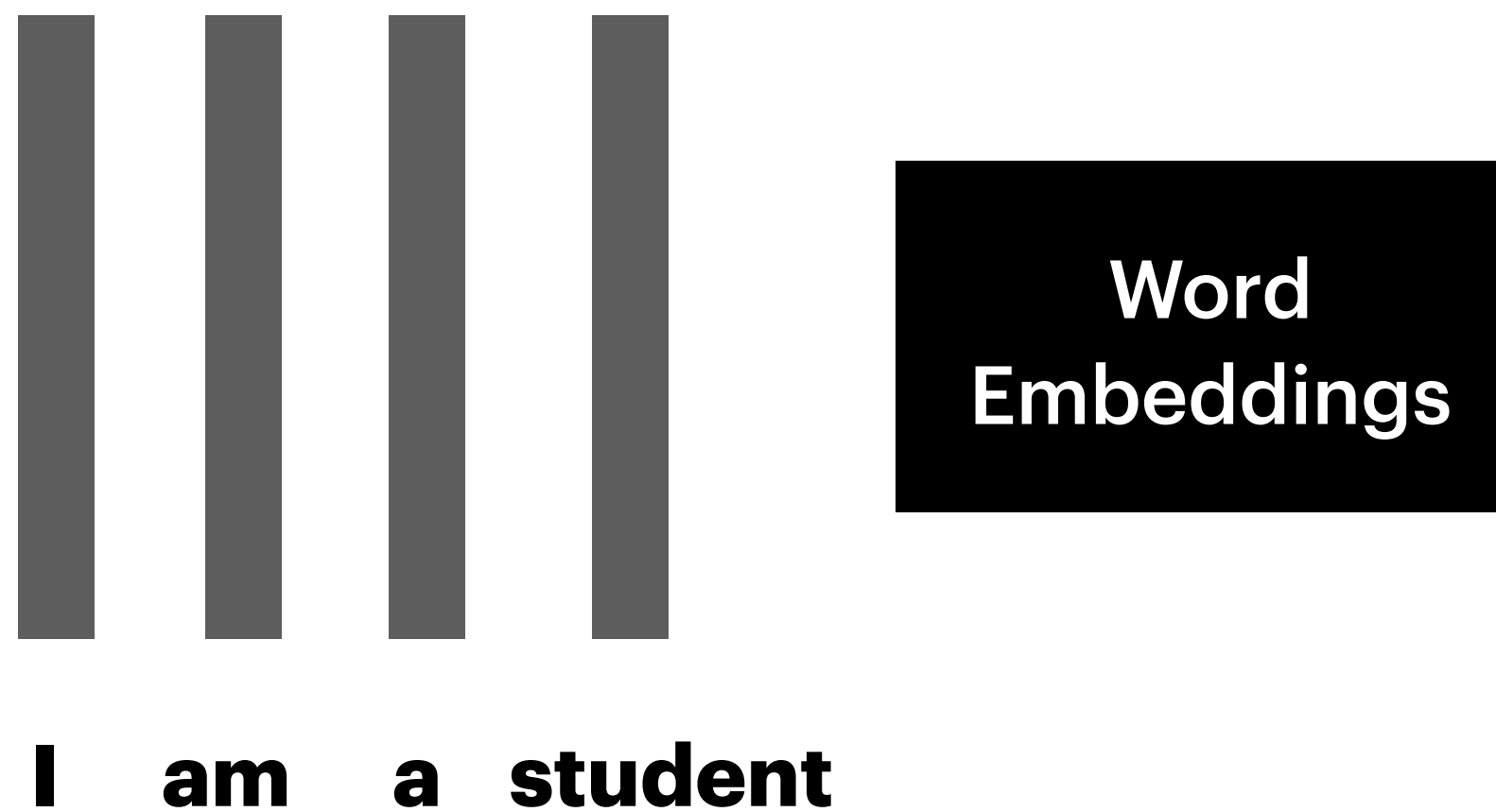
W_1

word2vec visualization



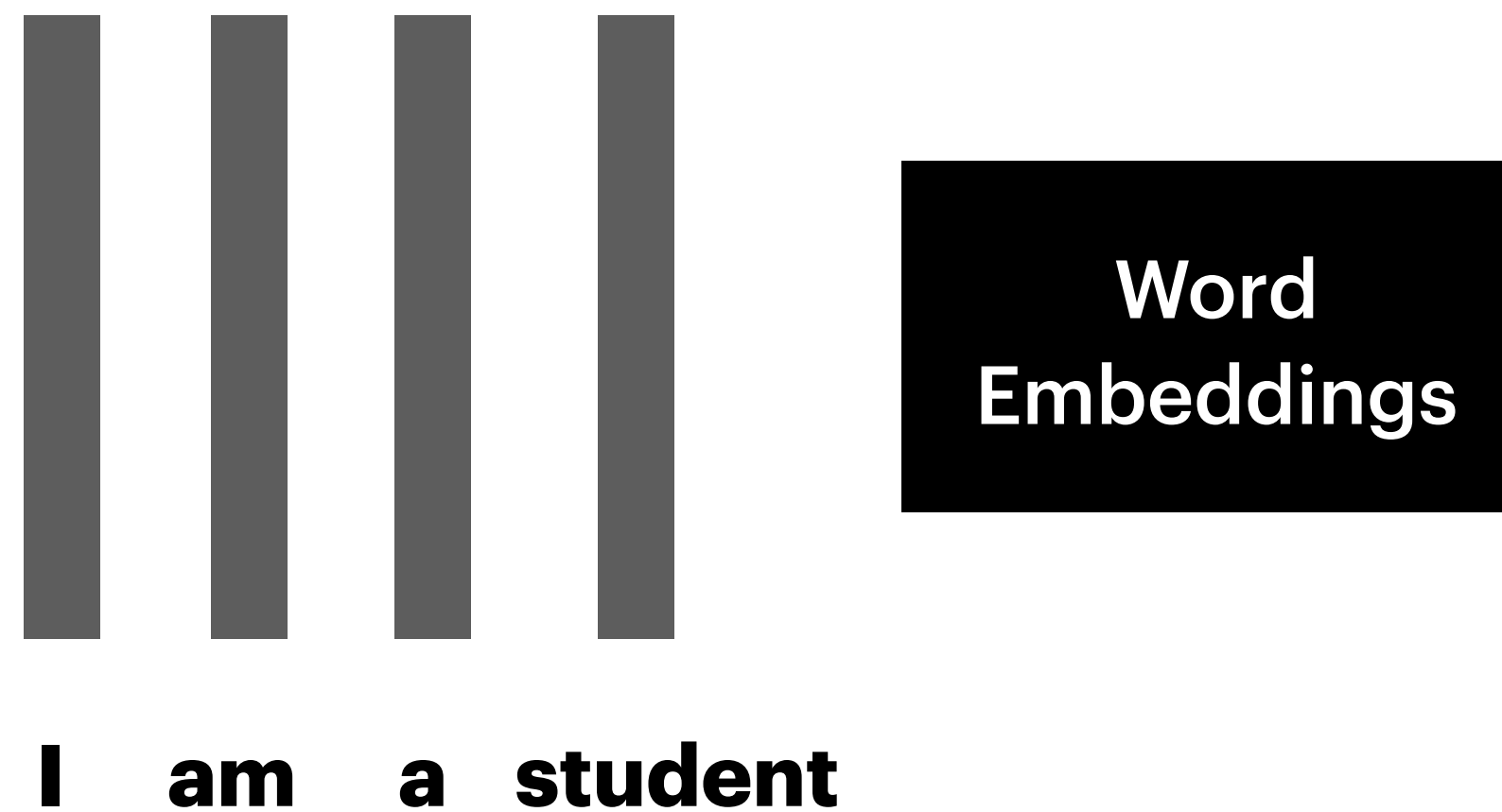
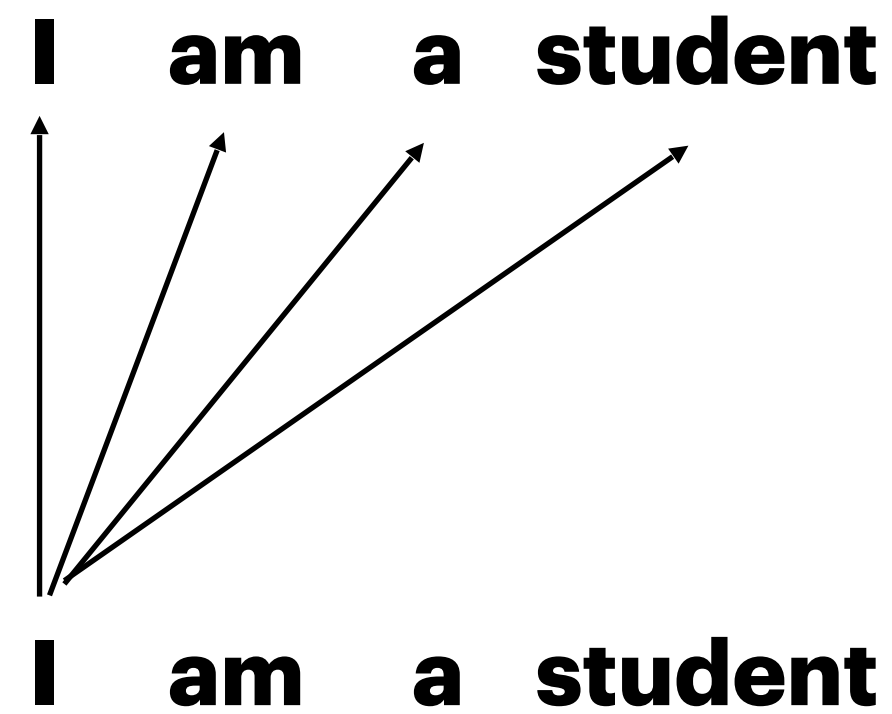
Transformers

- Embedding context in sequence inputs



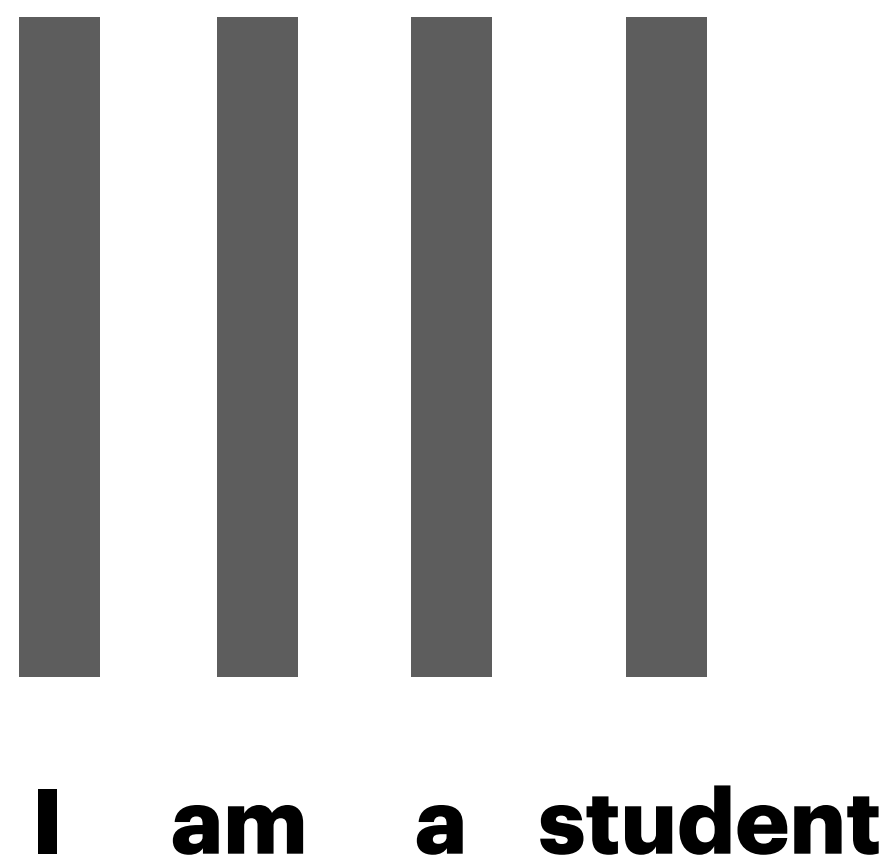
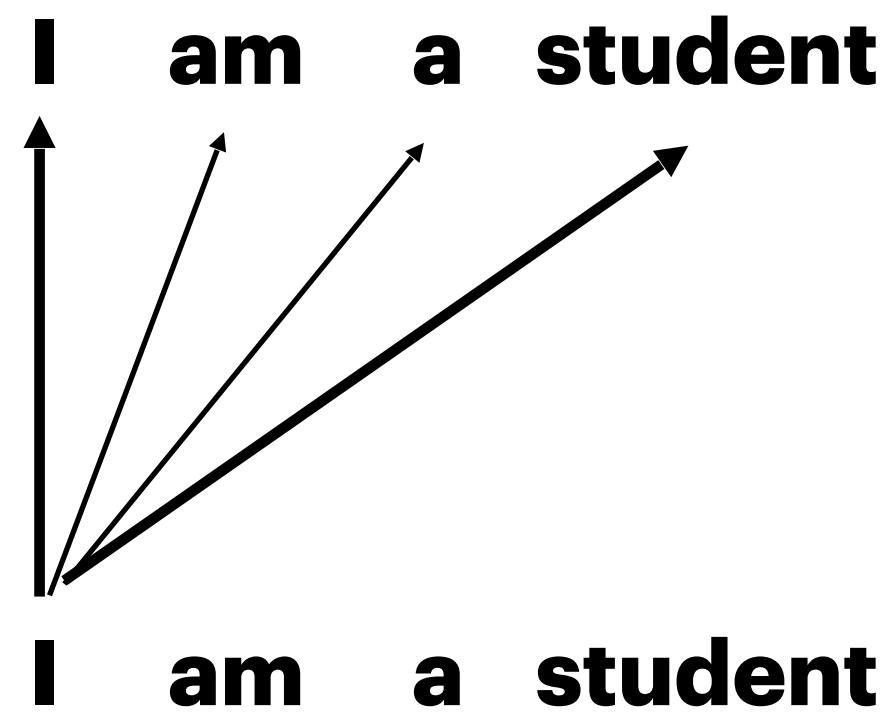
Transformers - self-attention

- Embedding context in sequence inputs
 - * Let us take an example



Transformers - self-attention

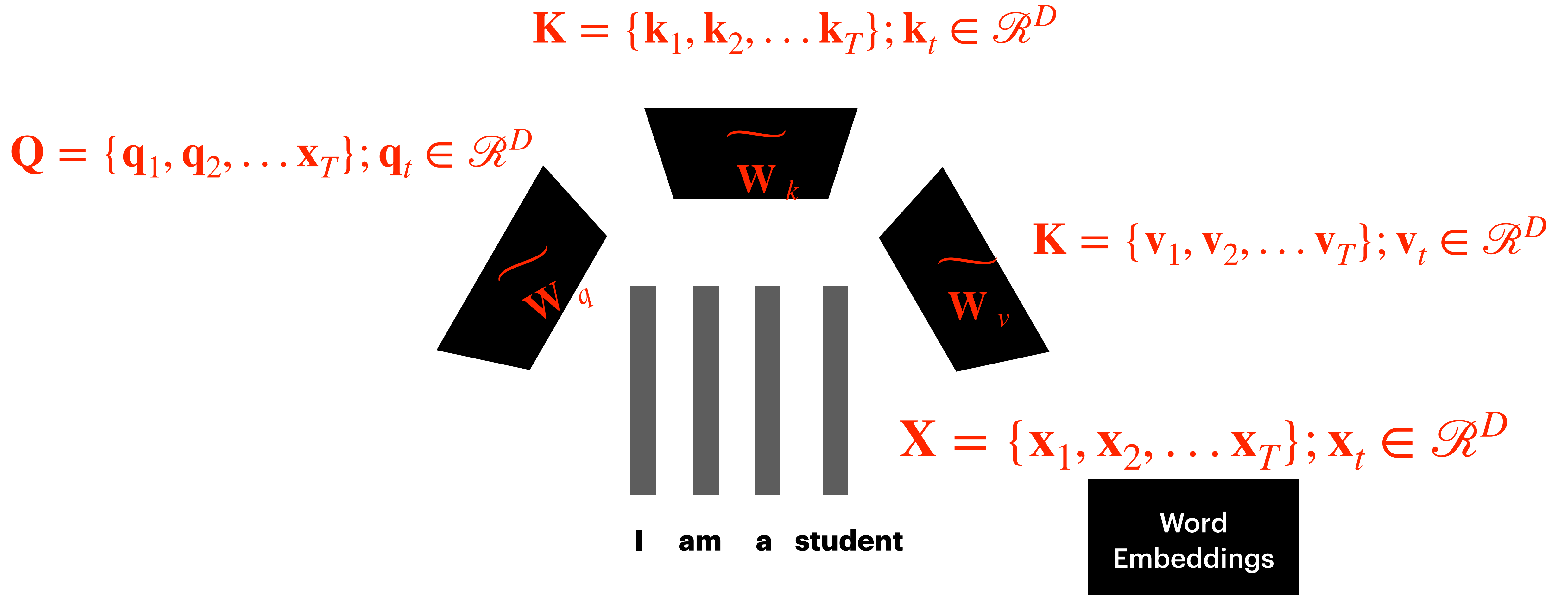
- Embedding context in sequence inputs
 - * Let us take an example
 - * Using word embeddings as the input representation



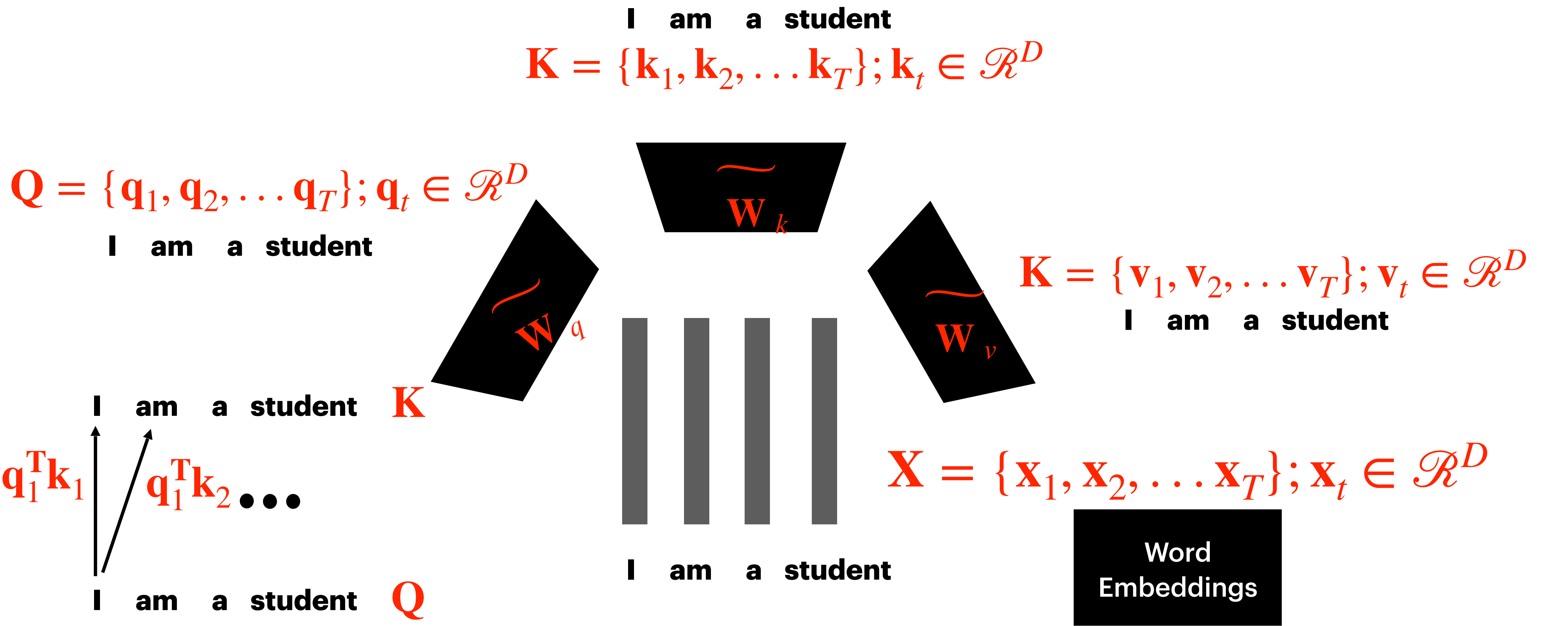
$$X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}; \mathbf{x}_t \in \mathcal{R}^D$$

Word
Embeddings

Transformers - self-attention



Transformers - self-attention



Transformers - self-attention

I am a student

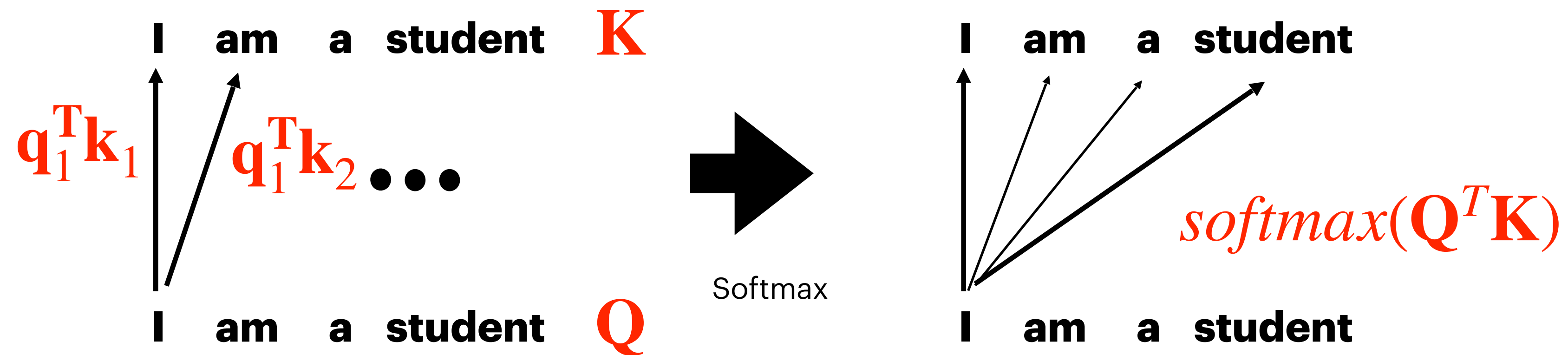
$$\mathbf{K} = \{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_T\}; \mathbf{k}_t \in \mathcal{R}^D$$

I am a student

$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_T\}; \mathbf{q}_t \in \mathcal{R}^D$$

$$\mathbf{K} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_T\}; \mathbf{v}_t \in \mathcal{R}^D$$

I am a student



Transformers - self-attention

I am a student

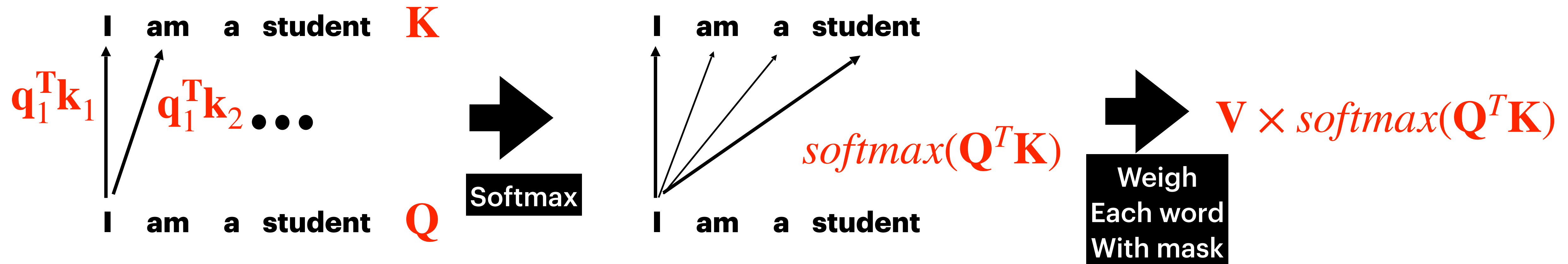
$$\mathbf{K} = \{\mathbf{k}_1, \mathbf{k}_2, \dots, \mathbf{k}_T\}; \mathbf{k}_t \in \mathcal{R}^D$$

I am a student

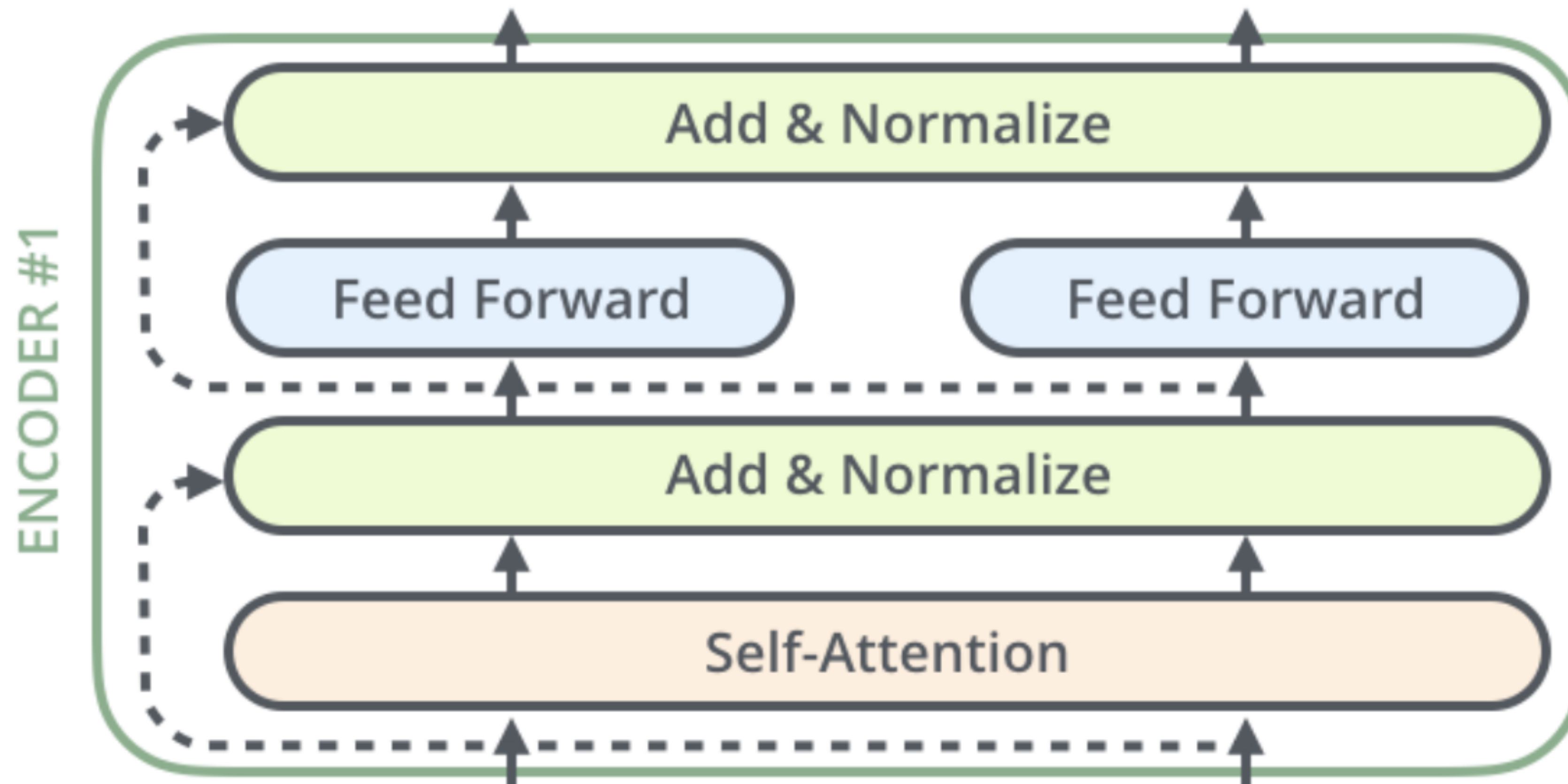
$$\mathbf{Q} = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_T\}; \mathbf{q}_t \in \mathcal{R}^D$$

$$\mathbf{K} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_T\}; \mathbf{v}_t \in \mathcal{R}^D$$

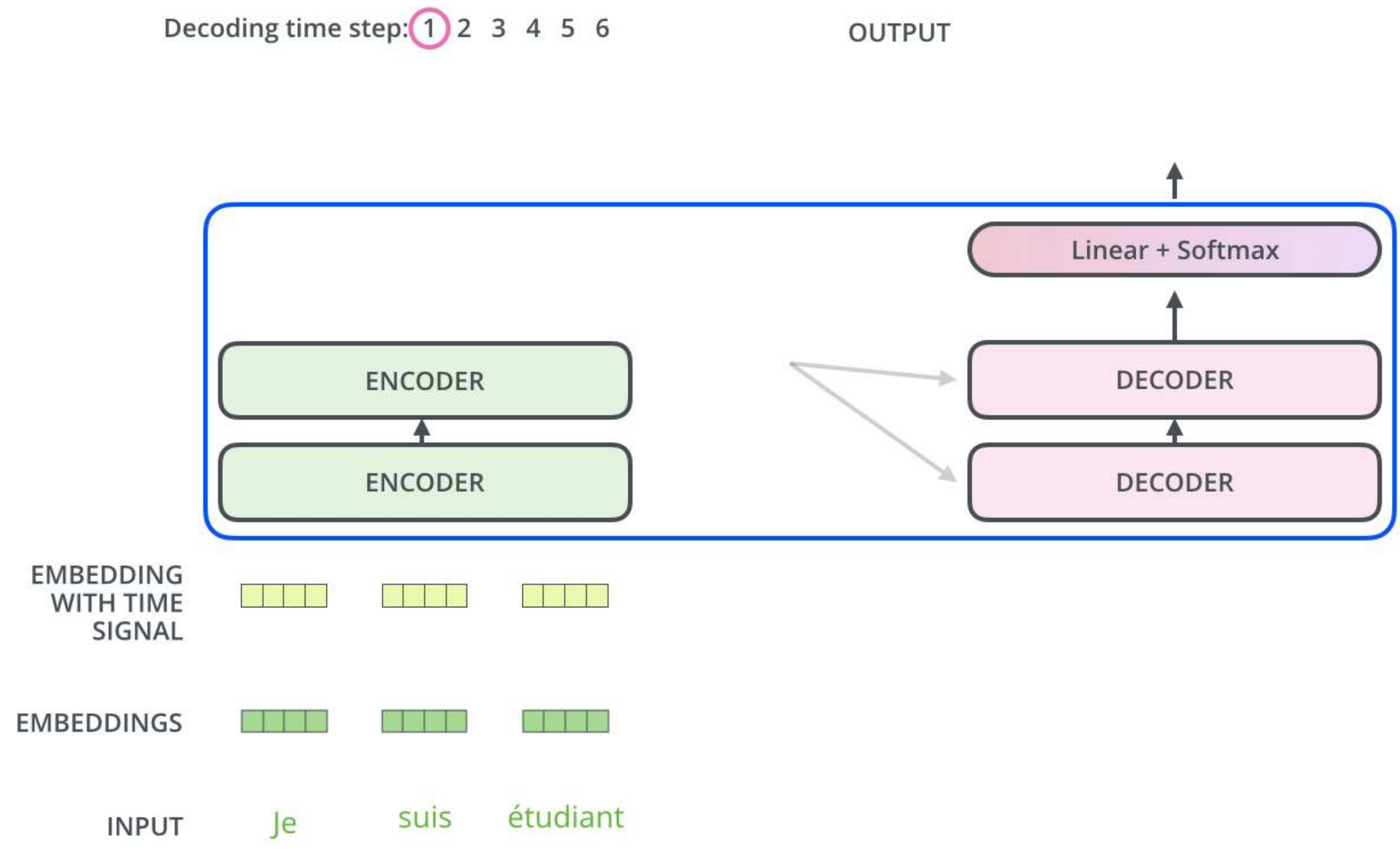
I am a student



Transformer encoder

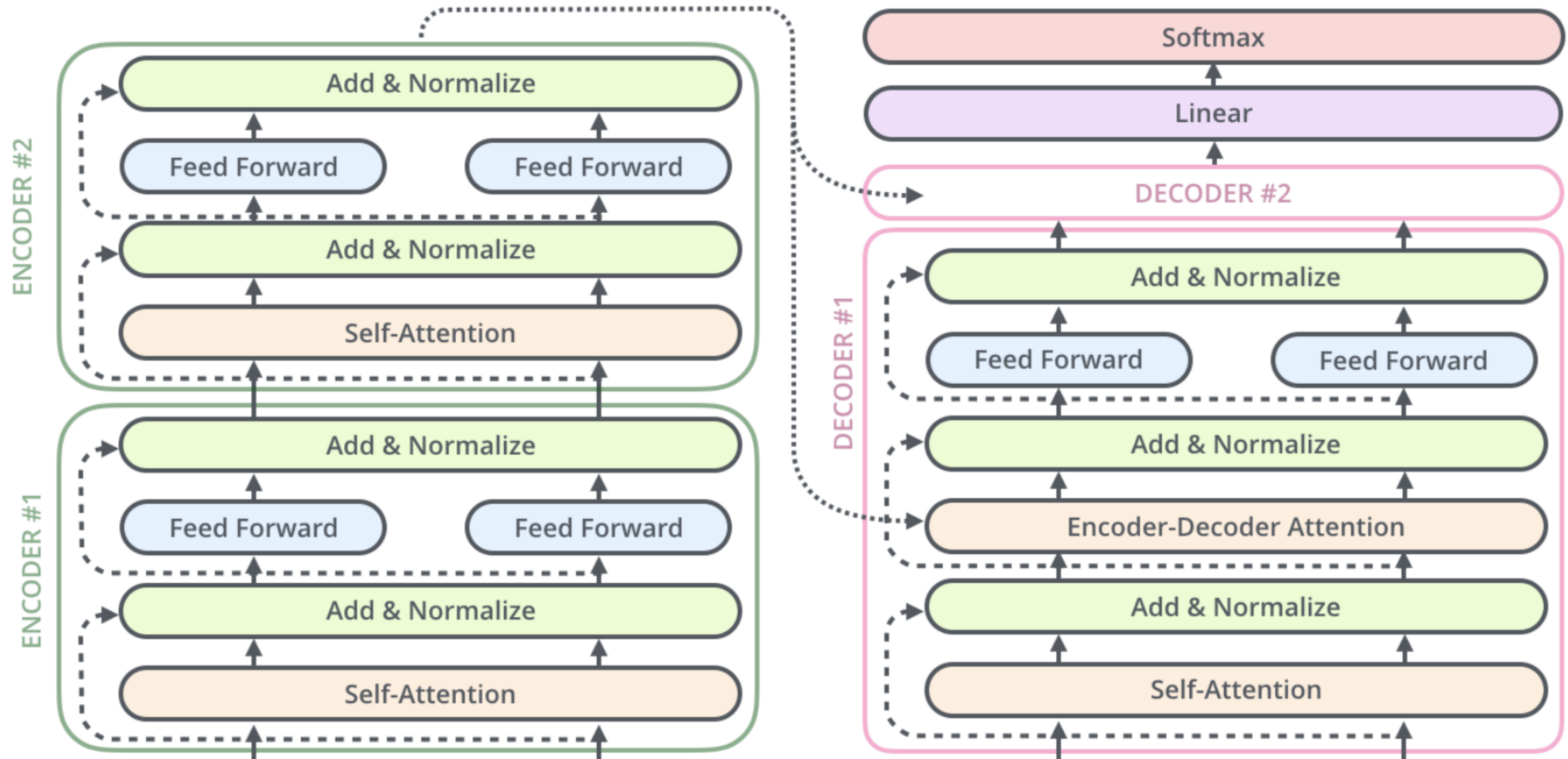


Transformer

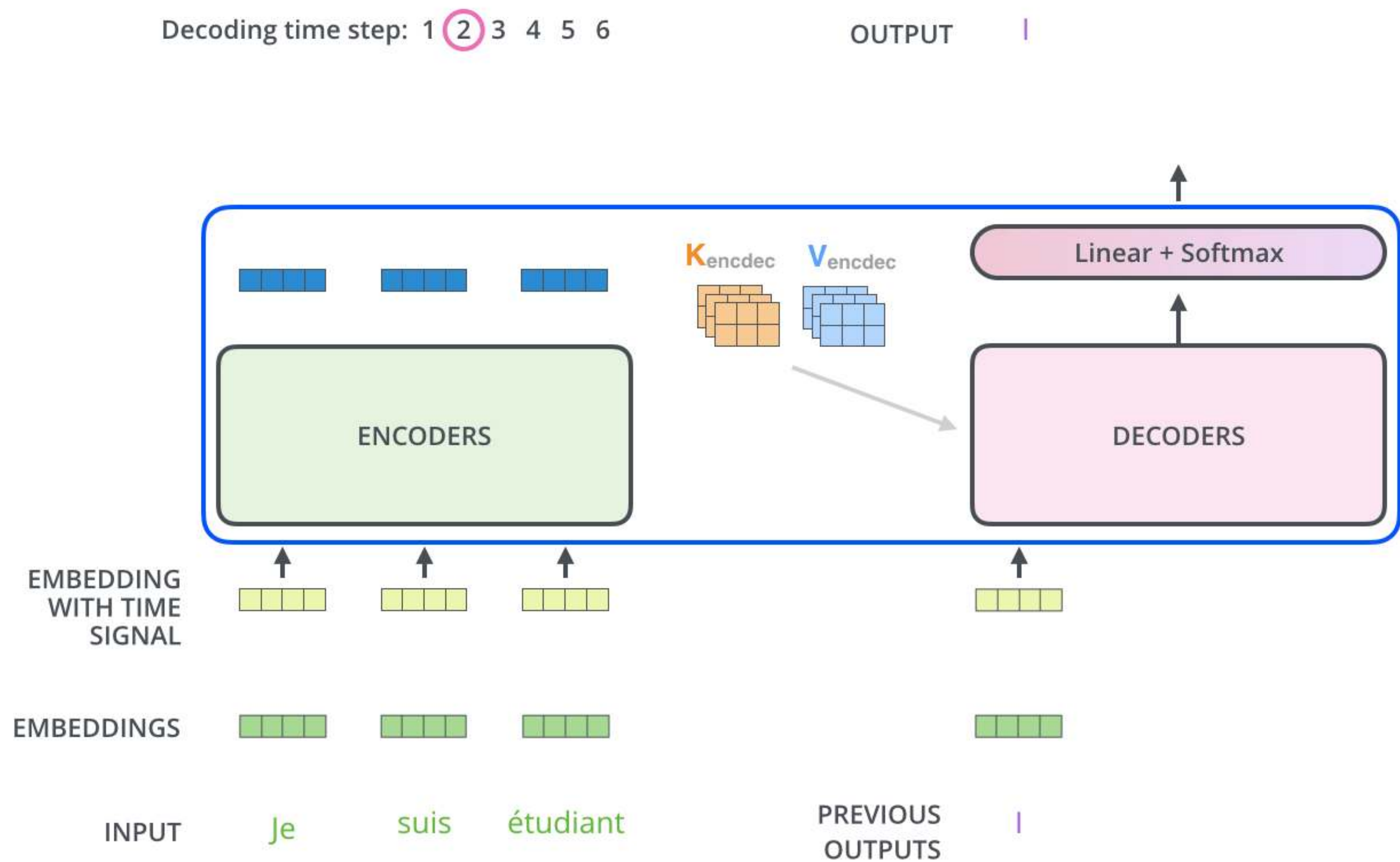


Pics taken from : <https://jalammar.github.io/illustrated-transformer/>

Transformer decoder



Transformer Example



THANK YOU

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